

Vacuum-sector mass gap for the local gauge-invariant sharp-local Yang–Mills net

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This paper proves a theorem about the bounded-region local gauge-invariant sharp-local net generated by flowed curvature composites for every compact connected simple gauge group G . The mass gap is proved for the vacuum Hamiltonian in the OS/Wightman reconstruction of that local net. Faithful-Wilson universality is theorem-level qualitative only: within the admissible gradient-flow renormalization class and after coherent normalization, different faithful finite-dimensional unitary Wilson regularizations yield canonically isomorphic continuum local theories. The explicit Route 1 quantitative certificate retained here remains indexed by the chosen faithful regularization ρ . Extended-support line and surface sectors, and more generally nonlocal sector data, lie outside the theorem-level claim. The core paper organizes the proof through a finite theorem-level reduction with one ultraviolet/public-scope gate, one Route 1 lattice-to-continuum vacuum-gap chain, one Lane A sharp-local construction chain, and one endpoint module; the detailed analytic proofs remain in Companions I–III.

I. MAIN THEOREM, SCOPE BOUNDARIES, AND EXACT ENDPOINT

This paper proves a vacuum-sector mass-gap theorem for the bounded-region local gauge-invariant sharp-local net generated by flowed curvature composites for every compact connected simple gauge group G . The mass gap concerns the vacuum Hamiltonian in the OS/Wightman reconstruction of that local net. Faithful-Wilson universality enters only qualitatively: within the admissible gradient-flow renormalization class and after coherent normalization, different faithful ultraviolet Wilson regularizations yield the same continuum local theory, while the explicit Route 1 quantitative certificate remains ρ -indexed. Extended-support line and surface sectors remain outside the theorem-level claim. We state those boundaries first because every later reduction in the paper is organized to preserve them exactly.

A. Main theorem and claim boundaries

Theorem I.1 (Main theorem). *Fix a compact connected simple Lie group G , choose a faithful finite-dimensional unitary Wilson regularization $\rho : G \rightarrow U(d_\rho)$, and fix a weak-window target coupling $u_{\text{ren}} \in (0, u_*]$ for the tuned dyadic construction. Let $\mathcal{A}_{\text{loc}}(G)$ denote the resulting bounded-region local gauge-invariant sharp-local net generated by flowed curvature composites, with exact theorem-level endpoint clarified in Section VII. Throughout the theorem-level group-only endpoint, faithful-Wilson identification is understood within the admissible gradient-flow renormalization class and after the coherent normalization conditions fixed in the endpoint packet. Then the following hold.*

(A) **One-shot entrance.** *There exist explicit constants and thresholds depending on (G, ρ) , including $q^*(G, \rho) \in (0, 1)$, a bounded physical entrance constant $\text{Cent}(G, \rho) < \infty$, and an entry index $N_{\text{ent}}(G, \rho; u_{\text{ren}})$, such that the exact one-shot blocked specification enters the KP/Dobrushin basin at a bounded physical scale and remains there along the tuned dyadic trajectory.*

(B) **Dense-core Euclidean clustering.** *On a dense positive-time local probe core the Euclidean two-point functions decay exponentially at rate*

$$m_*(G, \rho; u_{\text{ren}}) = \frac{m_{\text{blk}}(G, \rho)}{2 \text{Cent}(G, \rho) \ell_0}, \quad m_{\text{blk}}(G, \rho) := -\log q^*(G, \rho) > 0.$$

- (C) **Continuum local gauge-invariant Yang–Mills theory.** *There exists a unique continuum sharp-local state on $\mathcal{A}_{\text{loc}}(G)$ with Euclidean OS strength, Wightman/Haag–Kastler reconstruction, field correspondence for the renormalized gauge-invariant local differential polynomials produced by Lane A, and a non-triviality witness.*
- (D) **Vacuum mass gap.** *If $(\mathcal{H}_{\text{loc}}, \Omega_{\text{loc}}, H_{\text{loc}})$ denotes the OS/Wightman reconstruction of that sharp-local state, then*

$$\text{Spec}(H_{\text{loc}}) \subset \{0\} \cup [m_*(G, \rho; u_{\text{ren}}), \infty), \quad \ker(H_{\text{loc}}) = \mathbb{C}\Omega_{\text{loc}}.$$

The continuum local theory depends only on G at theorem level up to the canonical faithful-Wilson identification proved in Section VII, within the admissible gradient-flow renormalization class and after coherent normalization. The present explicit Route 1 quantitative certificate remains indexed by the chosen faithful regularization ρ .

The theorem-level object is deliberately local. Faithful-Wilson universality identifies the continuum local theory for faithful ultraviolet regularizations of the same compact simple group, but the present manuscript does not upgrade the explicit quantitative lower-bound ledger to a faithful- ρ -uniform statement. The endpoint likewise excludes extended-support line and surface sectors from the theorem claim itself.

Non-claim. The present manuscript does *not* claim a faithful- ρ -uniform explicit quantitative entrance or mass-gap bound. The continuum local theory is faithful-Wilson universal, but the explicit Route 1 quantitative certificate remains indexed by the chosen faithful regularization.

Non-claim. The present manuscript does *not* claim a vacuum gap for extended-support or other nonlocal sectors. The gap statement concerns only the vacuum Hamiltonian in the OS/Wightman reconstruction of the bounded-region local gauge-invariant sharp-local net.

Non-claim. The theorem-level group-only endpoint is asserted only within the admissible gradient-flow renormalization class and after coherent normalization. The present manuscript does *not* claim a scheme-independent identification outside that class or without those normalization conventions.

B. Imported standard framework inputs

The manuscript uses only three standard framework inputs as black-box imports, and they are kept explicit throughout the proof spine. First, after the Euclidean dossier has been

completed internally, the standard Osterwalder–Schrader reconstruction / uniqueness framework is used to pass from Euclidean Schwinger functions to the Hilbert-space / Hamiltonian setting. Second, at the final operator-theoretic readout, the spectral theorem is used to pass from the sharp-local semigroup bounds to the spectral-gap conclusion. Third, on the lattice side, standard Wilson-lattice reflection positivity is used as the usual positive-time positivity input. No broader external analytic package is hidden behind these names: the nonstandard Route 1, Lane A, and endpoint analysis is carried by the synchronized companion proof homes, and Appendix D records the same boundary for the Lean audit layer.

Representative background references used across the synchronized core-plus-companions package include the original Osterwalder–Schrader and lattice gauge sources^{1–4}; asymptotic-freedom, Wilson-flow, and lattice-improvement references^{5–7,23,24}; parabolic-kernel and compact-group harmonic-analysis references^{8,9,25}; cluster-expansion and Dobrushin references^{10–12}; special-function, Toeplitz-minor, total-positivity, and generalized-convexity sources used in the ultraviolet audit modules^{13–22}; standard AQFT and reconstruction references^{26–31}; and companion-local semialgebraic bookkeeping references³².

C. Effective gauge group and local-net endpoint

Definition I.2 (Effective gauge group determined by the ultraviolet action). Let G be a compact Lie group and let $\rho : G \rightarrow U(d_\rho)$ be the representation used to define the Wilson plaquette weight. Set

$$K_\rho := \ker(\rho) \subset Z(G), \quad G_{\text{eff}} := G/K_\rho.$$

The ultraviolet action, the Wilson lattice measure, and the strictly local gauge-invariant observables considered in this paper depend on (G, ρ) only through G_{eff} .

Lemma I.3 (Center-blindness and descent to the quotient). *With the Wilson action defined from ρ , the one-plaquette weight, the full Wilson lattice gauge measure, and every gauge-invariant observable generated by flowed local curvature composites depend on G only through the quotient G_{eff} . Equivalently, the resulting Schwinger functions and the reconstructed sharp-local OS/Wightman theory are canonically identified for (G, ρ) and $(G_{\text{eff}}, \rho_{\text{desc}})$.*

Proof. The plaquette action density has the form

$$A_\rho(U) = 1 - \frac{1}{d_\rho} \text{Re Tr}_\rho(U).$$

If $k \in K_\rho$, then $\rho(kU) = \rho(U)$ and therefore $A_\rho(kU) = A_\rho(U)$. The Wilson weight is constant on cosets of K_ρ , and the lattice measure is invariant under pointwise multiplication of link variables by elements of K_ρ . Any gauge-invariant observable built from ρ , including flowed local composites of the curvature, is likewise K_ρ -invariant, so its expectation depends only on the induced variables in G/K_ρ . Passing to the continuum limit and OS reconstruction preserves these identities. $\square$

Lemma I.4 (Adjoint-local generators and insensitivity to global form). *The sharp-local algebra $\mathcal{A}_{\text{loc}}(G)$ is generated by renormalized local composite insertions obtained as small-flow-time limits of flowed curvature composites. Every such generator is adjoint-local: it depends on the gauge field only through the associated adjoint bundle and is therefore invariant under central gauge transformations. Consequently, for any central subgroup $K \leq Z(G)$, the local net generated by $\mathcal{A}_{\text{loc}}$ is canonically identified for G and G/K .*

Proof. By construction, the flowed probe algebra is generated by gauge-invariant flowed curvature composites built from $G_{\mu\nu}(t, x)$ and its covariant derivatives. Lane A identifies each sharp-local insertion as a small-flow-time limit of inverse-flow combinations of these probes. Under a gauge transformation $g(x) \in G$, the flowed curvature transforms by the adjoint action, and the same holds for covariant derivatives. Since the defining polynomials are adjoint-invariant, every generator is unchanged under central gauge transformations. Because the adjoint action is identical for G and G/K , the resulting strictly local gauge-invariant net is canonically the same for both global forms. $\square$

D. What the paper does not claim

Corollary I.5 (Scope of the vacuum-gap statement). *The vacuum spectral gap asserted in Theorem I.1 is a statement about the OS vacuum representation of the bounded-region local gauge-invariant net generated by $\mathcal{A}_{\text{loc}}(G)$. In particular, it does not assert a gap in other representations or superselection sectors generated by nonlocal line or surface operators.*

Proof. This is an immediate consequence of Theorems I.3 and I.4. The theorem-level object is the strictly local gauge-invariant sharp-local net. Nonlocal defects and line/surface sectors are logically downstream extensions of that local theory, not part of the theorem-level endpoint proved here. $\square$

E. The three load-bearing theorem modules

A first-pass audit does not require a linear read of the full technical corpus. The proof burden concentrates in three theorem modules: the ultraviolet/public-scope gate of Section III; the Route 1 lattice-side gap and weak-window closure of Section IV; and the Lane A plus QE3 transport chain that closes the continuum sharp-local vacuum gap. The purpose of the next section is to make that reduction explicit at theorem level before any detailed proof reading begins.

With the claim and its boundaries fixed, the remaining task is to expose the proof as a finite theorem object. The next section therefore replaces the impression of a long monograph by an explicit public packet reduction, names the only live Route 1 and Lane A chains, and isolates the closure nodes that a first-pass referee must keep in view throughout the paper.

II. PUBLIC PROOF SPINE AND PACKET REDUCTION

The proof burden of Theorem I.1 is finite at theorem level. More precisely, the result reduces to ten public packets together with one auxiliary weak-window certificate, and every load-bearing step lies on one of two named chains: Route 1 for the mass-gap transport and Lane A for the sharp-local construction. This section records that reduction explicitly so that the rest of the paper can be read as a controlled packet stack rather than as a synchronized 368-page suite. Appendix A records the frozen live dependency DAG, and Appendix C records the master theorem index and canonical numbering crosswalk fixed in Phase I.

A. Ten-packet reduction and public theorem docket

Proposition II.1 (Packet reduction of the main theorem). *Theorem I.1 follows from ten public theorem packets together with one auxiliary certification theorem. More precisely:*

- (i) *Packets 1–2 imply part (A) of Theorem I.1.*
- (ii) *Packets 4–7 construct the continuum sharp-local state, the full inductive-union sharp-local algebra, and the bounded-base cyclicity needed for part (C).*
- (iii) *Packet 3 together with Packet 8, with the Lane A seam nodes consumed inside Packet 8, yield part (D).*

- (iv) *Packet 9 packages the downstream Euclidean clustering reformulation together with OS/Wightman existence, field correspondence, and non-triviality.*
- (v) *Packet 10 fixes the exact theorem-level local-net endpoint.*

The only additional certification theorem not on this public docket is the weak-window sufficiency certificate of Section IV.

Proof. This is a bookkeeping reduction. Each implication is exactly the package-level theorem statement named in Appendix A. No additional analytic input enters here. The auxiliary certificate records that every surviving weak-window use factors through the repaired curvature-sector package and that no hidden branch re-enters the public Route 1 chain. $\square$

B. Three first-pass theorem modules

A first-pass assessment can be bounded by three theorem modules.

Ultraviolet/public-scope gate.:

Packet 1 together with Packet 2. This module decides whether the compact-simple A1 input, public group-scope export, and one-shot entrance theorem are present on the public proof spine.

Route 1 lattice-side burden.:

Packet 3 together with the same-scale Wilson return, the Lane A seam nodes consumed in Packet 8, and the sole live weak-window certificate. This module decides whether a tuned full fixed-lattice lower bound is attached to the physical Wilson theory before the continuum transport begins.

Lane A and QE3 closure chain.:

Packets 4–8. This module decides whether the proof reaches the full sharp-local inductive union, preserves bounded-base cyclicity there, and transports the fixed-lattice lower bound to the continuum sharp-local vacuum sector.

C. Decisive seam nodes

After packet compression, the reader need only keep the following seam nodes explicitly in view.

Theorem F.331(i).:

Unique downstream mixed-channel closure node at finite cap.

Promoted bridge into Theorem 5.74.:

The theorem-facing positive/unital finite-cap bridge packages exactly what the capwise extension theorem may use from the bounded-family and mixed-channel closure package, and nothing more.

Theorem 5.74(b).:

Finite-cap sharp-local OS structure on the capwise algebra; this is the capwise OS input used before any inductive-union passage.

Corollary 5.75.:

Bounded-state restriction compatibility node on the QE3 seam; it is *not* the inductive-union theorem.

Corollary 5.76.:

Passage from capwise sharp-local reconstruction to the full inductive-union sharp-local algebra.

Theorem 5.77.:

First theorem-level bounded-base cyclicity export on the Lane A spine.

Lemma F.214.:

Final QE3 density/graph-core handoff; closes the cap/flow-time seam with no hidden domain argument.

Theorem F.216.:

Unique transport node carrying the tuned full fixed-lattice lower bound to the continuum sharp-local Hilbert space.

Theorem F.298.:

Sole live weak-window sufficiency certificate on the public Route 1 spine.

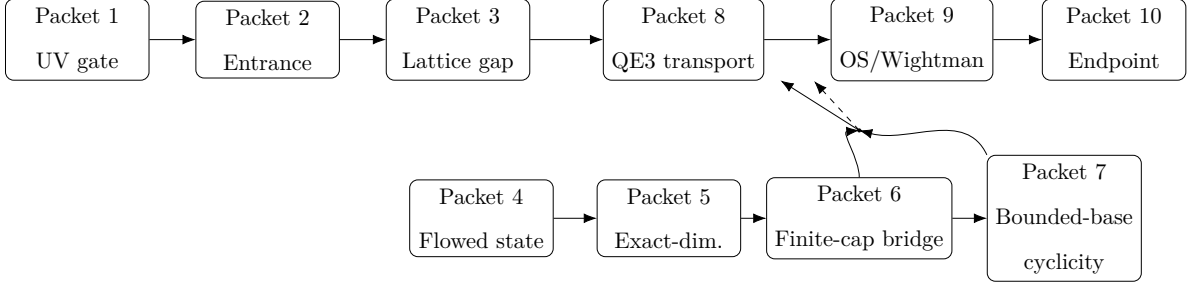


FIG. 1. Packet-flow schematic for the public theorem spine. The bounded first-pass mass-gap route is $\text{Packet 1} \Rightarrow \text{Packet 2} \Rightarrow \text{Packet 3} \Rightarrow \text{Packet 8}$, while the constructive Lane A chain runs through Packets 4–7 and feeds the seam data consumed in Packet 8. Packet 9 packages the downstream OS/Wightman and non-triviality consequences, and Packet 10 fixes the exact local-net endpoint.

D. Route 1 and Lane A reading order

The public theorem spine consists of two named load-bearing chains. At the claim-facing level the shortest mass-gap compression is

$$\text{Packet 1} \Rightarrow \text{Packet 2} \Rightarrow \text{Packet 3} \Rightarrow \text{Packet 8},$$

with the proviso that Packet 8 explicitly consumes the Lane A seam nodes Theorem 5.74(b), Corollary 5.75, Corollary 5.76, Theorem 5.77, Corollary F.213, and Lemma F.214, and still requires the weak-window sufficiency certificate of Theorem F.298. The unique constructive chain is

$$\text{Packet 4} \Rightarrow \text{Packet 5} \Rightarrow \text{Packet 6} \Rightarrow \text{Packet 7},$$

where the mixed-correlator closure node and the promoted finite-cap state bridge occur first, the capwise OS-structure theorem is still Theorem 5.74, the inductive-union passage occurs only afterward at Corollary 5.76, and bounded-base cyclicity appears only at the final theorem of the chain.

For a concentrated first pass the expensive checks are concentrated in three places only: the compact-simple ultraviolet/public-scope gate in Packet 1, the fixed-lattice-to-continuum transport seam across Packets 3 and 8, and the exact-dimension-to-inductive-union closure route across Packets 5–7. Everything else should be treated, on first pass, as certification, endpoint packaging, or explicitly deferred illustration.

Bounded verification path. A referee verifying only the vacuum-sector mass-gap statement may restrict attention to the finite theorem-facing route

$$\text{Packet 1} \Rightarrow \text{Packet 2} \Rightarrow \text{Packet 3} \Rightarrow \text{Packet 8},$$

together with the seam nodes Theorem F.331, Theorem 5.74(b), Corollary 5.75, Corollary 5.76, Theorem 5.77, Corollary F.213, Lemma F.214, Theorem F.216, and Theorem F.298. The companion corpus remains available for full attack and endpoint audit, but the theorem-facing route needed to reach part (D) of Theorem I.1 is already bounded at this level.

Lean formalization boundary. A Lean 4 formal companion is available as an independent audit layer for this theorem-facing proof spine. At its current submission boundary it packages, in an axiom-free manuscript-facing native object, the ten-packet reduction, the live dependency spine, the decisive seam nodes, the source-label crosswalk, and the named constructive, vacuum-gap, and endpoint witness exports. Its role is to make theorem routing, dependency locality, and closure of the manuscript-facing architecture mechanically checkable. It should not be read as a claim that every analytic proof-home argument has already been internalized natively in Lean. The standard framework inputs retained outside the current Lean boundary are exactly the Osterwalder–Schrader reconstruction / uniqueness framework, the spectral theorem used at the semigroup-to-gap step, and standard Wilson-lattice reflection positivity; Section D records this scope explicitly.

From this point onward the reader should track only three front-line modules: the ultraviolet/public-scope gate, the Route 1 fixed-lattice-to-continuum gap chain, and the Lane A sharp-local closure chain. Everything else is either a supporting proof home, a claim-facing endpoint packet, or deferred illustrative material. The remaining sections state those modules in that same order.

III. ULTRAVIOLET GATE AND ONE-SHOT ENTRANCE

The first principal module is the ultraviolet/public-scope gate. Its job is to place the compact-simple theorem on the stated class of groups, export the public group-scope input read later by Route 1, and close the one-shot entrance theorem at bounded physical scale. Once those three results are stated, every later quantitative use of the ultraviolet input can be treated as certified rather than re-derived.

A. Compact-simple A1 ultraviolet theorem

Theorem III.1 (Compact-simple A1 ultraviolet gate). *Let G be compact connected simple and let ρ be the chosen faithful finite-dimensional unitary Wilson regularization. Then there exist constants $c_{\text{A1}}(G, \rho) > 0$ and $\beta_{\text{A1}}(G, \rho) \geq 1$ such that for all $\beta \geq \beta_{\text{A1}}(G, \rho)$, all $\alpha \in [1/2, 1]$, and all nontrivial irreducible representations π occurring in the character expansion of the Wilson plaquette weight,*

$$|a_\pi(\beta, \alpha)| \leq \exp\left(-c_{\text{A1}}(G, \rho) \frac{C_2(\pi)}{\beta + \sqrt{C_2(\pi)}}\right).$$

Equivalently, after descent to $G_{\text{eff}} = G/\ker(\rho)$ when necessary, the one-plaquette Fourier coefficients satisfy a crossover decay bound that is valid on the full stated class of compact connected simple groups.

B. Public group-scope export

Corollary III.2 (Public group-scope export). *For every compact connected simple group G and every chosen faithful Wilson regularization ρ , the ultraviolet constants furnished by Theorem III.1 are available on the public Route 1 spine. In particular, the one-shot entrance theorem and the faithful-Wilson endpoint theorems may be read on the full stated class of compact connected simple groups without any separate exceptional-family proviso.*

Proof. This is the public export furnished by Component 1. Once the compact-simple A1 theorem is proved on the stated class, the later Route 1 and endpoint resultss consume only the resulting ultraviolet constants and the fact that the theorem holds on the full compact-simple scope. $\square$

C. One-shot entrance into the KP/Dobrushin basin

Proposition III.3 (One-shot entrance at bounded physical scale). *Let $B(\beta; G, \rho)$ denote the explicit one-shot entrance block size furnished by the Route 1 construction. Then there exist constants $\beta^*(G, \rho) > 0$, $q^*(G, \rho) \in (0, 1)$, $\text{Cent}(G, \rho) < \infty$, and $N_{\text{ent}}(G, \rho; u_{\text{ren}}) \in \mathbb{N}$ such that the following hold.*

- (a) *For every bare coupling $\beta \geq \beta^*(G, \rho)$, the exact one-shot blocked specification at scale $B(\beta; G, \rho)$ lies in the KP/Dobrushin basin with contraction bounded by $q^*(G, \rho)$, uniformly in finite volume and admissible boundary conditions.*
- (b) *Along the tuned dyadic trajectory $a_n = \ell_0 2^{-n}$, $\beta_n = \beta(a_n; u_{\text{ren}}, G, \rho)$, the physical entrance scale is bounded:*

$$a_n B(\beta_n; G, \rho) \leq \text{Cent}(G, \rho) \ell_0 \quad (n \geq N_{\text{ent}}(G, \rho; u_{\text{ren}})).$$

The output of this section is exactly what Route 1 consumes: a compact-simple A1 theorem, a public group-scope export, and a one-shot entrance theorem. The long harmonic-analysis and comparison proofs are moved to Companion I; the core paper keeps only the theorem-level front end needed to continue to the fixed-lattice gap.

IV. ROUTE 1 LATTICE-SIDE GAP AND WEAK-WINDOW CLOSURE

This section isolates the lattice-side content of the mass-gap proof. The goal is not to replay Appendix F in miniature, but to state the three theorem-level facts that must be in hand before the continuum transport can begin: a tuned full fixed-lattice OS gap, a same-scale return to the physical Wilson theory, and a certificate that the repaired curvature-sector weak-window certificate is the only downstream weak-window input.

A. Tuned full fixed-lattice OS gap

Proposition IV.1 (Tuned full fixed-lattice OS gap). *Let $a_n = \ell_0 2^{-n}$ and $\beta_n = \beta(a_n; u_{\text{ren}}, G, \rho)$ be the tuned dyadic trajectory determined by the one-shot entrance data of Proposition III.3. Define*

$$m_{\text{blk}}(G, \rho) := -\log q^*(G, \rho), \quad m_*(G, \rho; u_{\text{ren}}) := \frac{m_{\text{blk}}(G, \rho)}{2 \text{Cent}(G, \rho) \ell_0}.$$

Then the full fixed-lattice OS Hamiltonian gap along the tuned sequence satisfies

$$\liminf_{n \rightarrow \infty} \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq m_*(G, \rho; u_{\text{ren}}) > 0.$$

Equivalently, the lattice-side Route 1 proof yields a tuned-sequence full fixed-lattice lower bound that is already normalized at the physical scale required later by the continuum transport.

B. Same-scale return to the physical Wilson theory

Theorem IV.2 (Same-scale Wilson-to-patched transport). *Fix a support budget $M_{\text{supp}} \in \mathbb{N}$ and a depth-growth constant $C_{\text{depth}} > 0$. For bounded cylinder observables supported in at most M_{supp} unit coarse blocks, let $E_{a,m}^W$ denote the exact Wilson evaluation obtained by the m -step blocked recursion and let $E_{a,m}^{\text{good}}$ denote the parallel chart-truncated evaluation. Then for every integer $N \geq 1$ there exist $u_0(N, C_{\text{depth}}, M_{\text{supp}}) > 0$ and $C_N(C_{\text{depth}}, M_{\text{supp}}) < \infty$ such that, whenever $u(a) \leq u_0$ and the dyadic depth satisfies*

$$1 \leq m \leq C_{\text{depth}} \log(1/u(a)),$$

one has

$$|E_{a,m}^W(H) - E_{a,m}^{\text{good}}(H)| \leq C_N \|H\|_{L^\infty} u(a)^N,$$

uniformly in the cutoff a , uniformly in all such depths m , and uniformly over the finite-volume/admissible-boundary-condition class used later in Route 1. In particular the same estimate applies at every fixed physical comparison scale $L = 2^m a$, since then $m = O(\log(1/u(a)))$ on the asymptotically free window. The same estimate extends to fixed-scale covariances for bounded cylinder pairs whose product remains in the same fixed-scale test class (equivalently, whose common support hull occupies at most M_{supp} unit coarse blocks), and to flowed local observables after flow localization and the finite-depth local transport closure.

Dependency summary.

Inputs. exact Wilson one-step block factorization, the nested blocked Wilson recursion, the averaged good-event transport estimate, and the finite-depth local transport closure on the fixed-scale Route 1 test class

Imported standard background. none

Conclusion at this node. same-scale Wilson and chart-truncated evaluations agree up to $O(u(a)^N)$ uniformly for bounded-support observables and logarithmic dyadic depths $m = O(\log(1/u(a)))$; the same comparison holds for the fixed-scale covariance class and for flowed local observables after localization

Used later in. Theorem IV.4, Theorem VI.1, and Theorem VI.9

Remark IV.3 (Frozen depth regime on the Wilson-return theorem). The uniformity in Theorem IV.2 is exactly the logarithmic-depth window stated in its detailed proof: dyadic depths $m \leq C_{\text{depth}} \log(1/u(a))$. This is the regime later used for one-shot entrance, fixed

physical comparison scales, and the bounded Wilson/dyadic OS limit. The theorem should therefore neither be paraphrased as a merely fixed-depth estimate nor overread as uniform for arbitrary depth families.

C. Weak-window closure and transport summary

Theorem IV.4 (Weak-window sufficiency certificate). *All live downstream weak-window uses in the manuscript factor through the repaired curvature-sector package WW_{curv} and through nothing stronger. More precisely:*

1. *the only weak-window export to the dyadic beta-remainder analysis is the sign input $b_0 > 0$;*
2. *the one-shot entrance-scale comparison uses only the repaired package listed in the weak-window definition;*
3. *the tuned-sequence physical entrance corollary uses only that repaired package together with the explicit one-shot comparison theorem;*
4. *the one-shot entrance theorem itself does not use any weak-window convexity statement beyond the repaired package; and*
5. *consequently the Route 1 chain — one-shot entrance, tuned fixed-lattice gap, and continuum vacuum-gap transport — contains no hidden dependence on any edge-Hessian convexity hypothesis.*

Remark IV.5 (Route 1 transport summary). For theorem-level use, the fixed-lattice-to-continuum gap route is frozen as follows:

$$\begin{aligned} &\text{Corollary N.21} \Rightarrow \text{Proposition F.3} \Rightarrow \text{Proposition F.4} \Rightarrow \text{Theorem IV.2} \Rightarrow \text{Theorem} \\ &\text{F.211} \Rightarrow \text{Theorem F.212} \Rightarrow \text{Theorem F.216} \Rightarrow \text{Corollary F.5.} \end{aligned}$$

In the completed core-paper renumbering this becomes the public Route 1 chain from the group-scope export and one-shot entrance theorem through the tuned fixed-lattice gap, the same-scale Wilson return, the QE3 transport theorem, and the final continuum-gap corollary. The only live weak-window input on that chain is Theorem IV.4, and Lane B remains downstream only.

At this stage the proof has a fixed-lattice lower bound attached to the physical Wilson theory and protected by a sole live weak-window certificate. No continuum conclusion is claimed yet. The next section turns to the only constructive route that carries this input into the full sharp-local algebra: Lane A.

V. LANE A: FINITE-CAP SHARP-LOCAL CONSTRUCTION AND INDUCTIVE-UNION CLOSURE

Lane A is the only constructive route used in the proof from the flowed continuum state to the full sharp-local inductive-union algebra. Its internal logic must therefore stay literal in the core paper: exact-dimension quotient control feeds one-shell transport and finite-truncation inverse control; that package feeds the first mixed-correlator bridge at finite cap and then the theorem-level positive/unital finite-cap bridge into the capwise OS theorem; only afterward does the proof pass from capwise sharp-local states to the full inductive union and finally export bounded-base cyclicity. The section is long because those role boundaries are themselves part of the proof architecture.

Scope discipline. Every Lane A statement below is to be read with its exact scope frozen. The finite-cap theorems concern one fixed engineering cap at a time, and any constants, graph norms, or invariant cores may depend on that cap. Corollary V.11 is only the qualitative passage to the inductive union. Theorem V.14 is the first full sharp-local density export. No theorem in this section claims a quantitative estimate uniform over the entire inductive union unless that uniformity is stated explicitly.

A. Flowed continuum state and bounded positive-time base state

Theorem V.1 (Unique flowed continuum state). *Fix a target renormalized coupling value $u(\mu) = u_\mu$ in the asymptotically free interval and, for each lattice spacing a , tune the ultraviolet coupling so that this renormalization condition holds. Then the chart-truncated dyadic RG outputs converge without subsequences on $\mathcal{A}_{\text{flow}}$, and the physical Wilson expectations converge to the same limit. More precisely, there exists a unique Euclidean-invariant state ω_{flow} on $\mathcal{A}_{\text{flow}}$ such that for every $F \in \mathcal{A}_{\text{flow}}$,*

$$\omega_{\text{flow}}(F) = \lim_{a \downarrow 0} \omega_a(F) = \lim_{a \downarrow 0} \omega_a^{\text{RG}}(F).$$

The limit is independent of ultraviolet irrelevant initialization once the renormalized coupling trajectory has been fixed.

Corollary V.2 (Tuned bounded positive-time base state). *Fix $u_{\text{ren}} \in (0, u_*]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}})$$

be the associated tuned dyadic trajectory. Then for every finite family $\mathcal{F} \subset A_+$ of bounded cylinder observables supported in $\{x_0 > 0\}$, the Wilson expectations $\{\omega_{a_n}(F)\}_{F \in \mathcal{F}}$ converge jointly as $n \rightarrow \infty$. In particular, for every $F \in A_+$ the limit

$$\omega_{\text{bd}}(F) := \lim_{n \rightarrow \infty} \omega_{a_n}(F)$$

exists and defines a state on the bounded positive-time algebra.

Proof. This is exactly the tuned bounded positive-time base-state corollary in Companion II, Appendix A (source Corollary 4.94). It uses the same no-subsequence convergence and observable-evaluation Lipschitz estimate for finite bounded families in A_+ , but it does not require the flowed de-patching seam. $\square$

B. Exact-dimension quotient blocks

Theorem V.3 (Exact-dimension quotient identity). *For every exact-dimension symmetry block $V_{\Delta, \epsilon, \lambda}$ in the graded local-operator quotient, the descended small-flow substitution acts as the identity on the associated-graded block:*

$$\text{gr}_{\Delta}(\Gamma_t) \upharpoonright_{V_{\Delta, \epsilon, \lambda}} = I_{V_{\Delta, \epsilon, \lambda}}.$$

Equivalently, the tree-level small-flow matching on each exact-dimension quotient block is basis-free and exactly the identity.

Theorem V.4 (Coefficient extraction on arbitrary quotient blocks). *Fix an exact-dimension symmetry block $V_{\Delta, \epsilon, \lambda}$ and choose the canonical local and flowed bases attached to that block. Let $C_{\Delta, \epsilon, \lambda}^{\text{UV, tree}}$ be the tree-level exact-dimension diagonal block of the ultraviolet matching matrix obtained by coefficient extraction after small-flow substitution. Then*

$$C_{\Delta, \epsilon, \lambda}^{\text{UV, tree}} = I.$$

In particular, the exact-dimension diagonal ultraviolet matching problem on an arbitrary symmetry block is an intrinsic finite-dimensional linear-algebra problem attached to the quotient block itself, and the same matrix is seen in every engineering cap containing that block.

C. One-shell transport and finite-truncation inverse control

Theorem V.5 (One-shell transport on finite truncations). *Fix an exact-dimension symmetry block $s = (\Delta, \epsilon, \lambda)$ with canonical local basis, canonical flowed basis, and canonical block norm. Let $Y_{s,j}(f)$ denote the vector of canonical flowed lifts at dyadic times $t_j = 2^{-2j}t_0$, ordered in the fixed canonical basis. Then there exist scale-independent constants $C_s, C_{s,f} < \infty$ and canonically defined linear maps*

$$A_{s,j} : V_s \rightarrow V_s, \quad B_{ss',j} : V_{s'} \rightarrow V_s \quad (s' \prec s),$$

such that

$$Y_{s,j+1}(f) = (I_s + u_j A_{s,j}) Y_{s,j}(f) + \sum_{s' \prec_s} u_j B_{ss',j} Y_{s',j}(f) + Q_{s,j}(f),$$

with

$$\|A_{s,j}\|_{\text{op,can}} + \sum_{s' \prec_s} \|B_{ss',j}\|_{\text{op,can}} \leq C_s, \quad \|Q_{s,j}(f)\|_{L^2(\|\cdot\|_{\text{can}})} \leq C_{s,f} u_j^2.$$

Equivalently, after quotienting by lower blocks of the grading order, exact-dimension one-shell transport is $I_s + u_j A_{s,j} + O(u_j^2)$.

Theorem V.6 (Finite-truncation inverse control). *Fix $\Delta_{\max} \geq 0$ and order the canonical symmetry blocks of engineering dimension at most $\Delta_{\max}$ by increasing engineering dimension and then by a fixed symmetry-type order. Let*

$$C^{(\leq \Delta_{\max})}(t; \mu)$$

be the full coefficient matrix from the canonical flowed lifts of that truncation to the corresponding renormalized local basis. Then, for all sufficiently small t :

- (i) $C^{(\leq \Delta_{\max})}(t; \mu)$ is block lower triangular in the chosen grading order;
- (ii) each diagonal block satisfies the canonical polylogarithmic inverse bound;
- (iii) each off-diagonal block is uniformly bounded on $(0, t_0]$ by the insertion-integration and insertion-to-OS bounds together with finite-depth composition in the fixed truncation.

Consequently there exist constants $C_{\Delta_{\max}, \text{inv}} < \infty$ and $p_{\Delta_{\max}} < \infty$ such that

$$\left\| \left(C^{(\leq \Delta_{\max})}(t; \mu) \right)^{-1} \right\|_{\text{op}} \leq C_{\Delta_{\max}, \text{inv}} \left(\frac{u(\mu)}{u(\mu_t)} \right)^{p_{\Delta_{\max}}} \quad (0 < t \leq t_0),$$

so the quantitative inverse-control hypothesis holds on every finite engineering-dimension truncation.

Dependency summary.

Inputs. the exact-dimension quotient package of Theorems V.3 and V.4, the one-shell transport theorem V.5, and the insertion-integration / insertion-to-OS bounds on each fixed truncation recorded in the detailed proof

Imported standard background. none

Conclusion at this node. every fixed engineering cap has a canonical block-lower-triangular coefficient matrix with a polylogarithmic inverse bound; no full-inductive-union uniformity is asserted

Used later in. Corollary V.7, Theorem V.9, and Theorem V.10

Corollary V.7 (Finite-truncation SFTE/remainder package). *Fix $\Delta_{\max} \geq 0$ and the canonical blockwise basis. Then the canonical flowed lifts and the renormalized local fields on that truncation satisfy a small-flow-time expansion*

$$Y_t^{(\leq \Delta_{\max})}(f) = C^{(\leq \Delta_{\max})}(t; \mu) O_{\text{ren}}^{(\leq \Delta_{\max})}(f; \mu) + R_t^{(\leq \Delta_{\max})}(f; \mu)$$

with block-lower-triangular coefficient matrix. There exists $\eta_{\Delta_{\max}} > 0$ such that every fixed capwise vacuum or mixed channel of the remainder is $O(t^{\eta_{\Delta_{\max}}})$, hence compatible with any polylogarithmic inverse bound. Equivalently, on every fixed finite cap the centered inverse-flow approximants are Cauchy as $t \downarrow 0$.

Proof. This is the capwise remainder package produced by the canonical insertion-RG decomposition together with the finite-truncation inverse control of Theorem V.6. The compatibility with polylogarithmic inverse factors is exactly what allows the finite-cap closure theorem to be stated without any extra sectorwise hypotheses. $\square$

The explicit scalar, tensor, and loop sectors are retained only as illustrations and normalization cross-checks in Companion II. They are not the essential widening step in the core paper. What matters here is that Theorems V.6 and V.7 provide unconditional finite-truncation inverse and remainder control on every canonical engineering cap.

D. Mixed-correlator bridge at finite cap

Corollary V.8 (Finite mixed bounded-family packaging). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$.*

Let

$$B_{\text{cap}}^{(\leq \Delta_{\text{max}})} := * \left(A_{\text{flow}}^{(\leq \Delta_{\text{max}})}, A_+ \right)$$

be the bounded capwise algebra generated by the capwise flowed subalgebra and the bounded positive-time algebra. For every finite family $W \subset B_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ there is a unique packaged state

$$\omega_{W, \text{cap}, \text{bnd}}^{(u_{\text{ren}})}$$

on the finitely generated algebra $\text{Alg}(W)$ such that:

- (i) *tuned Wilson expectations of every finite family in $\text{Alg}(W)$ converge jointly along the associated dyadic trajectory;*
- (ii) *the packaged state is positive, unital, and agrees with ω_{flow} on $\text{Alg}(W) \cap A_{\text{flow}}^{(\leq \Delta_{\text{max}})}$ and with ω_{bd} on $\text{Alg}(W) \cap A_+$;*
- (iii) *the packaged states are compatible on overlaps and inherit the time-translation covariance later used at the mixed-channel seam.*

Proof. The finite-family algebra generated by a fixed bounded set is still a bounded finite observable set, so the same convergence, positivity, and overlap-compatibility arguments used earlier for the flowed and bounded positive-time states apply after packaging one finite family at a time. $\square$

Theorem V.9 (First mixed-correlator closure at finite cap). *Fix $u_{\text{ren}} \in (0, u_*]$ and $\Delta_{\text{max}} \geq 0$.*

Let $A_{\text{loc}}^{(\leq \Delta_{\text{max}})\wedge}$ be the abstract unital $$ -algebra generated by $B_{\text{cap}}^{(\leq \Delta_{\text{max}})}$ together with the formal centered capwise local symbols*

$$\widehat{O}_j^{(\leq \Delta_{\text{max}})}(f; \mu).$$

For every finite family of bounded words fixing the ambient mixed channel, the centered inverse-flow representatives are Cauchy as all flow parameters tend to 0, and the resulting limits define a unique positive unital functional on $A_{\text{loc}}^{(\leq \Delta_{\text{max}})\wedge}$. Equivalently, once the capwise state has been defined, every formal centered symbol is realized by the corresponding centered renormalized local insertion, and the mixed-channel limit is independent of the relative rates at which the local flow parameters tend to 0.

Dependency summary.

Inputs. Corollary V.8, Corollary V.7, Theorem V.6, and the fixed-cap telescoping reduction of mixed channels used in the detailed proof

Imported standard background. none

Conclusion at this node. for one fixed engineering cap and one fixed mixed channel, the centered inverse-flow representatives converge to a unique positive unital capwise functional on the abstract mixed algebra; the statement does not yet pass to the inductive union and does not yet assert bounded-base cyclicity

Used later in. Theorem V.10

At the detailed proof, the next step is now theorem-level: Companion II promotes an explicit finite-cap state bridge into source Theorem 5.74 that packages exactly the positive/unital capwise functional extracted from the bounded-family and mixed-channel closure package. The same detailed proof now isolates the finite-cap positivity mechanism as a separate lemma: reflection positivity is read only through bounded positive-time representatives, and the inverse-flow replacement error tends to zero before the capwise OS theorem is invoked. The capwise OS-structure theorem then uses only that local package; the inductive-union passage and bounded-base cyclicity remain separate later exports.

Companion II, Section 7 also records the state-construction summary: tuned bounded Wilson base state on A_+ , finite bounded-family packaging, mixed-correlator closure on one fixed cap, the promoted positive/unital capwise bridge, the capwise OS field realization, the inductive-union state, and finally bounded-base cyclicity. No later QE3 or endpoint theorem uses any additional functional-backbone step outside that summary.

E. Finite-cap sharp-local extension

Theorem V.10 (Finite-cap sharp-local extension). *Fix a weak-window target value $u_{\text{ren}} \in (0, u_*]$ and an engineering cutoff $\Delta_{\text{max}} \geq 0$. Let $A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ be the $*$ -algebra generated by the capwise flowed subalgebra, the bounded positive-time algebra, and the canonical renormalized local generators of engineering dimension at most Δ_{max} . Then Theorems V.6, V.7 and V.9 define a unique capwise sharp-local continuum state*

$$\omega_{\leq \Delta_{\text{max}}}^{(u_{\text{ren}})}$$

on $A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ such that:

- (a) *for every capwise basis index j and test function f , the centered inverse-flow approximants converge in every mixed capwise channel to the centered renormalized local field;*
- (b) *the Schwinger functions of these capwise renormalized local fields satisfy the Osterwalder–Schrader axioms, and the corresponding field polynomials are realized on the explicit common invariant core of bounded flowed positive-time classes;*
- (c) *the restriction of $\omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}$ to the capwise flowed subalgebra equals ω_{flow} , while its restriction to A_+ equals ω_{bd} and carries the same dyadic OS matrix elements.*

Dependency summary.

Inputs. Theorem V.9, Theorem V.6, Corollary V.7, and the theorem-level finite-cap state bridge together with the dedicated positivity lemma isolated in the Companion II detailed proof

Imported standard background. none beyond the standard OS reconstruction step on each finite cap, applied only after the finite-cap reflection-positive state has been constructed theorem-locally in the detailed proof

Conclusion at this node. one fixed engineering cap acquires a unique sharp-local state with capwise OS structure and bounded-state compatibility on the bounded subalgebras; the theorem does not yet assert the inductive-union passage or bounded-base cyclicity

Used later in. Corollary V.11 and Theorem VI.2

F. Inductive union and bounded-base cyclicity

Corollary V.11 (Passage to the full sharp-local inductive union). *For each $\Delta_{\max} \geq 0$, the finite truncation $A_{\text{loc}}^{(\leq \Delta_{\max})}$ satisfies the hypotheses of Theorem V.10. The resulting capwise sharp-local states are compatible under the natural inclusions*

$$A_{\text{loc}}^{(\leq \Delta_{\max})} \hookrightarrow A_{\text{loc}}^{(\leq \Delta'_{\max})} \quad (\Delta_{\max} \leq \Delta'_{\max}),$$

and therefore define a unique state on the inductive union

$$\mathcal{A}_{\text{loc}} := \varinjlim_{\Delta_{\max} \rightarrow \infty} A_{\text{loc}}^{(\leq \Delta_{\max})}.$$

This state extends ω_{bd} on A_+ and satisfies the OS axioms on the full sharp-local algebra generated by flowed curvature composites.

Proof. For each engineering cutoff $\Delta_{\max}$, Theorem V.10 furnishes a capwise sharp-local state

$$\omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}$$

on $A_{\text{loc}}^{(\leq \Delta_{\max})}$. Fix $\Delta_{\max} \leq \Delta'_{\max}$. We first prove the overlap identity on the smaller cap by reducing to the three generator classes that span it.

Step 1: overlap words in the bounded and flowed sectors. If W is a word built only from bounded positive-time observables and capwise flowed observables lying in $A_{\text{loc}}^{(\leq \Delta_{\max})}$, then Theorem V.10 part (c) identifies the restrictions of both capwise states with the same bounded base state ω_{bd} on A_+ and with the same flowed state ω_{flow} on the capwise flowed algebra. Hence

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W) = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W)$$

for every such overlap word W .

Step 2: overlap words containing renormalized local generators. Let

$$W = Q(B_1, \dots, B_r, O_1, \dots, O_s)$$

be a noncommutative word in which each B_i belongs to the overlap of the bounded or flowed sectors and each O_j is a renormalized local generator whose quotient label already appears below the smaller cap. The coherent normalization scheme fixes those quotient labels and normalized generators before the larger cap is chosen. For each O_j choose the inverse-flow approximants $O_{j,t}$ supplied by Theorem V.10; because the generator is already present in the smaller cap, the same approximating family $O_{j,t}$ is admissible in both caps. Set

$$W_t := Q(B_1, \dots, B_r, O_{1,t}, \dots, O_{s,t}).$$

Then W_t lies in the overlap algebra generated by the bounded and flowed sectors, so Step 1 gives

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W_t) = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W_t) \quad (t > 0).$$

Now Theorem V.10 part (a) identifies each O_j as the mixed-channel limit of $O_{j,t}$ in both caps. Passing to the limit $t \downarrow 0$ therefore yields

$$\omega_{\leq \Delta'_{\max}}^{(u_{\text{ren}})}(W) = \omega_{\leq \Delta_{\max}}^{(u_{\text{ren}})}(W)$$

for every overlap word W in the smaller cap.

Step 3: state equality on the overlap algebra. Finite linear combinations of overlap words span $A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$. By Steps 1–2 the two capwise states agree on that spanning set, hence

$$\omega_{\leq \Delta'_{\text{max}}}^{(u_{\text{ren}})} \upharpoonright_{A_{\text{loc}}^{(\leq \Delta_{\text{max}})}} = \omega_{\leq \Delta_{\text{max}}}^{(u_{\text{ren}})}.$$

Thus the finite-cap states are compatible under the natural inclusions.

Now let $A_{\text{loc}}^{\text{alg}} := \bigcup_{\Delta_{\text{max}}} A_{\text{loc}}^{(\leq \Delta_{\text{max}})}$ with the inductive-union algebra structure. Every algebra element lies in some finite cap, so define its expectation by evaluating it in any cap containing it. The compatibility just proved makes this cap-independent, and positivity / unitality are inherited capwise. This yields a well-defined state on

$$\mathcal{A}_{\text{loc}} = \varinjlim_{\Delta_{\text{max}} \rightarrow \infty} A_{\text{loc}}^{(\leq \Delta_{\text{max}})}.$$

Finally, every OS test polynomial involves only finitely many generators and therefore lies in one finite cap. For that polynomial, the Euclidean invariance, symmetry, and reflection-positivity clauses are exactly the clauses already proved for the corresponding capwise state in Theorem V.10, so the OS axioms pass verbatim to the inductive union. The restriction to A_+ is the same bounded state on every cap and therefore equals ω_{bd} on the full sharp-local algebra. $\square$

Dependency summary.

Inputs. the capwise states of Theorem V.10 together with the explicit overlap comparison on shared generators and inverse-flow representatives fixed by the coherent normalization scheme

Imported standard background. none

Conclusion at this node. the compatible finite-cap states glue to one qualitative sharp-local inductive-union state extending the bounded base state; this is only the passage to the full union and does not itself prove bounded-base cyclicity

Used later in. Theorem V.14, Theorem VI.2, and Theorem VII.1

Remark V.12 (Quantifier boundary at the inductive-union passage). Corollary V.11 is qualitative glueing only. It identifies the compatible finite-cap states on overlaps and defines the inductive-union sharp-local state, but it does not promote the capwise inverse-control, remainder, or graph-norm bounds to a statement uniform over the full union.

Remark V.13 (Where the lower bound is not read at the finite-cap seam). The capwise inverse bounds and graph norms in Section V are construction-local only: they are used

to build the finite-cap sharp-local states, to compare them on overlaps, and to export the bounded-base dense core. The quantitative lower bound m_* is *not* read from any family of capwise inverse constants, and no statement in this section asserts a full-union estimate uniform in $\Delta_{\max}$. The theorem-level role of Corollary V.11 and Theorem V.14 is precisely to make the full sharp-local state and the bounded-base dense core available before the spectral readout occurs later in Section VI. Thus the finite-cap to full-union passage is qualitative on the state side, while the quantitative vacuum lower bound is transported only afterward on the full sharp-local Hilbert space.

Theorem V.14 (Bounded-base cyclicity in the reconstructed sharp-local Hilbert space). *Fix $u_{\text{ren}} \in (0, u_*]$ and let ω_{sharp} be the full sharp-local state of Corollary V.11. Let $(H_{\text{loc}}, \Omega_{\text{loc}}, \mathcal{H}_{\text{loc}})$ be its OS reconstruction, and let $\mathcal{P}_{+, \text{sharp}}$ be the $*$ -algebra of finite positive-time polynomials in the renormalized local generators with test functions supported in $\{x_0 > 0\}$. Then:*

(i) *for every $P \in \mathcal{P}_{+, \text{sharp}}$ there exists a family $P_t^{\text{bnd}} \in A_+$ such that*

$$\|[P_t^{\text{bnd}}] - [P]\|_{\mathcal{H}_{\text{loc}}} \longrightarrow 0 \quad (t \downarrow 0);$$

(ii) *the image of A_+ is dense in $\mathcal{H}_{\text{loc}}$, equivalently $\Omega_{\text{loc}} = [1]$ is cyclic for the bounded positive-time base algebra inside the full sharp-local reconstruction;*

(iii) *the centered bounded classes satisfy*

$$\overline{\text{span}}\{[F - \omega_{\text{sharp}}(F)] : F \in A_+\} = \Omega_{\text{loc}}^\perp.$$

Proof. Fix $P \in \mathcal{P}_{+, \text{sharp}}$. Only finitely many renormalized local generators occur in P , so there exists an engineering cap $\Delta_{\max}$ such that $P \in A_{\text{loc}}^{(\leq \Delta_{\max})}$. Write

$$P = Q(O_1, \dots, O_r)$$

for a fixed noncommutative polynomial Q in capwise generators $O_1, \dots, O_r$. For each O_ν , choose a sequence of RP-safe bounded positive-time representatives

$$B_{\nu, n} \in A_+, \quad n \in \mathbb{N},$$

obtained from the inverse-flow representatives in Theorem V.10 along any fixed sequence $t_n \downarrow 0$, so that

$$\|[B_{\nu, n}] - [O_\nu]\|_{\mathcal{H}_{\text{loc}}} \longrightarrow 0 \quad (n \rightarrow \infty)$$

for each ν . Define the bounded approximating sequence

$$P_n^{\text{bnd}} := Q(B_{1,n}, \dots, B_{r,n}) \in A_+.$$

Because the cap is fixed and Q has finite degree, every factor appearing in Q acts on the same finite-cap invariant core, and the remaining bounded factors stay uniformly controlled in the finitely many capwise graph norms relevant to P . Expanding the noncommutative polynomial difference by the usual telescoping identity gives

$$P_n^{\text{bnd}} - P = \sum_{\nu=1}^r T_{\nu,n} (B_{\nu,n} - O_\nu),$$

where each $T_{\nu,n}$ is a finite linear combination of monomials in the other factors. The capwise bounds just noted therefore give a constant $C_P < \infty$, depending only on the chosen cap and on P , such that

$$\|[P_n^{\text{bnd}}] - [P]\|_{\mathcal{H}_{\text{loc}}} \leq C_P \sum_{\nu=1}^r \|[B_{\nu,n}] - [O_\nu]\|_{\mathcal{H}_{\text{loc}}}.$$

Since the right-hand side tends to 0, we obtain

$$\|[P_n^{\text{bnd}}] - [P]\|_{\mathcal{H}_{\text{loc}}} \longrightarrow 0 \quad (n \rightarrow \infty),$$

which proves part (i).

To prove part (ii), fix $\psi \in \mathcal{H}_{\text{loc}}$ and $\varepsilon > 0$. Choose a Cauchy sequence of positive-time sharp-local polynomials $P^{(m)} \in \mathcal{P}_{+, \text{sharp}}$ whose OS classes converge to ψ . Pick m so large that

$$\|\psi - [P^{(m)}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon/2.$$

Applying part (i) to the single polynomial $P^{(m)}$, choose n such that

$$\|[P_n^{\text{bnd}}] - [P^{(m)}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon/2.$$

Then $P_n^{\text{bnd}} \in A_+$ and

$$\|\psi - [P_n^{\text{bnd}}]\|_{\mathcal{H}_{\text{loc}}} < \varepsilon.$$

Because ψ and ε were arbitrary, the image of A_+ is dense in $\mathcal{H}_{\text{loc}}$. This proves cyclicity of the bounded positive-time base algebra.

For part (iii), every bounded class decomposes as

$$[F] = [F - \omega_{\text{sharp}}(F)] + \omega_{\text{sharp}}(F) \Omega_{\text{loc}}.$$

Thus the closed linear span of Ω_{loc} together with the centered bounded classes contains the dense image of A_+ and is therefore all of $\mathcal{H}_{\text{loc}}$. Each centered bounded class is orthogonal to Ω_{loc} , so their closed linear span is contained in $\Omega_{\text{loc}}^\perp$. Conversely, if $\eta \in \Omega_{\text{loc}}^\perp$, approximate η by bounded classes $[F_n]$ using part (ii) and replace each $[F_n]$ by the centered class $[F_n - \omega_{\text{sharp}}(F_n)]$; orthogonality to Ω_{loc} removes only the vacuum component. Hence

$$\overline{\text{span}}\{[F - \omega_{\text{sharp}}(F)] : F \in A_+\} = \Omega_{\text{loc}}^\perp,$$

which is exactly part (iii). □

Dependency summary.

Inputs. the finite-cap inverse-flow representatives and RP-safe bounded approximants supplied by Theorem V.10, together with the full sharp-local state of Corollary V.11

Imported standard background. none

Conclusion at this node. the bounded positive-time algebra is dense in the reconstructed full sharp-local Hilbert space and the closed linear span of its centered classes is the vacuum-orthogonal sector; no new gap statement is proved here

Used later in. Corollary VI.6 and Theorem VII.1

Remark V.15 (First full-sharp-local export on Lane A). Theorem V.14 is the first theorem that exports a full sharp-local density statement. In later QE3 uses every sharp-local polynomial vector is still handled one finite engineering cap at a time before this density conclusion is invoked. No full-union familywise graph-norm estimate is being asserted here.

The output of Lane A is now fixed in theorem-level form: a finite-cap sharp-local extension, an inductive-union passage to the full sharp-local algebra, and bounded-base cyclicity in the reconstructed sharp-local Hilbert space. Those are exactly the inputs later consumed at the QE3 seam and in the claim-facing endpoint theorem package; everything else in the larger manuscript that supports Lane A has been pushed to Companion II.

VI. QE3 SEAM AND CONTINUUM VACUUM-GAP TRANSPORT

The final transport step carries the tuned fixed-lattice lower bound to the continuum sharp-local vacuum sector without changing its theorem-level meaning. The key point is that this step has a single theorem-level interface package: the bounded Wilson limit, the

continuum-time OS upgrade, the density and graph-core handoff, and the transport theorem that reads the same lower bound in the full sharp-local Hilbert space. This section states that package exactly and nothing more.

A. Route 1 transport summary

On the public spine the fixed-lattice-to-continuum transport chain is

$$\begin{aligned} & \textit{Corollary III.2} \Rightarrow \textit{Proposition III.3} \Rightarrow \textit{Proposition IV.1} \Rightarrow \textit{Theorem IV.2} \\ & \Rightarrow \textit{Theorem VI.1} \Rightarrow \textit{Theorem VI.2} \Rightarrow \textit{Theorem VI.9} \Rightarrow \textit{Corollary VI.11}. \end{aligned}$$

The only live weak-window input on this chain is Theorem IV.4; no auxiliary edge-Hessian placeholder or other weak-window branch re-enters after that theorem. The first Lane A compatibility input consumed here is the bounded-state restriction clause isolated in the detailed proof as Source Corollary 5.75; the finite-cap sharp-local OS structure used before any passage to the union is the capwise statement of Source Theorem 5.74(b), and Corollary V.11 is only the later inductive-union passage. Lane A supplies the sharp-local OS reconstruction; Route 1 supplies the fixed-lattice lower bound; and Lane B remains downstream only.

Route discipline. The live QE3 route is now theorem-level in four explicit steps only: Theorem VI.1 records bounded dyadic OS limits on A_+ and nothing more; Theorem VI.2 is the unique upgrade from those dyadic limits to the all- t sharp-local OS semigroup; Corollary VI.6 together with Lemma VI.7 supply the only density and finite-graph-norm package used later; and Theorem VI.9 performs the actual gap transfer. Equivalently, on the QE3 side the first theorem that can change the status of the gap statement is Theorem VI.9; every earlier node only identifies the state, semigroup, or approximation package through which the same lattice-side lower bound is later read. No direct lattice-level all-time theorem and no hidden domain-completion step occur on the public spine.

B. Bounded Wilson/dyadic OS limit

Theorem VI.1 (Bounded Wilson/dyadic OS limit theorem). *Fix a weak-window target value $u_{\text{ren}} \in (0, u_*]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}}).$$

Define the dense dyadic translation set

$$D_{\text{dyad}} := \{m\ell_0 2^{-k} : m, k \in \mathbb{N}_0\}.$$

Then along this tuned dyadic sequence:

- (i) for every bounded cylinder observable $F \in A_+$, the limit

$$\omega_{\text{bd}}(F) := \lim_{n \rightarrow \infty} \omega_{a_n}(F)$$

exists and defines a state on A_+ ;

- (ii) for every bounded $F, G \in A_+$ and every dyadic shift $s \in D_{\text{dyad}}$, the OS matrix-element limit

$$K_{u_{\text{ren}}}(F, G; s) := \lim_{n \rightarrow \infty} \omega_{a_n}(\Theta F \tau_s G)$$

exists, and in particular $K_{u_{\text{ren}}}(F, F; 0) \geq 0$, so ω_{bd} is reflection positive on A_+ .

Dependency summary.

Inputs. the tuned bounded-base-state convergence package underlying Corollary V.2, the dyadic reflected matrix-element convergence proved in the Route 1 detailed proof, and Wilson reflection positivity at each lattice spacing

Imported standard background. none

Conclusion at this node. along the tuned dyadic trajectory the bounded Wilson state on A_+ and its dyadic reflected OS matrix elements converge; the theorem does not yet produce an all-time semigroup on the sharp-local Hilbert space

Used later in. Theorem VI.2 and Theorem VI.9

C. Continuum-time OS upgrade

Theorem VI.2 (Continuum-time OS upgrade of the bounded base state). *Fix $u_{\text{ren}} \in (0, u_*]$ and let ω_{sharp} be the full sharp-local state furnished by Corollary V.11 over the same tuned bounded base state. Let $(H, \Omega, \mathcal{H})$ be its OS reconstruction. Then:*

- (i) the restriction of ω_{sharp} to A_+ is exactly ω_{bd} ;
- (ii) for every bounded $F, G \in A_+$ and every dyadic shift $s \in D_{\text{dyad}}$,

$$\omega_{\text{sharp}}(\Theta F \tau_s G) = K_{u_{\text{ren}}}(F, G; s);$$

(iii) for every bounded $F, G \in A_+$ and every $t \geq 0$,

$$\omega_{\text{sharp}}(\Theta F \tau_t G) = \langle [F], e^{-tH} [G] \rangle_{\mathcal{H}},$$

and the map $t \mapsto \omega_{\text{sharp}}(\Theta F \tau_t G)$ is continuous on $[0, \infty)$.

Consequently the bounded positive-time state of Theorem VI.1 has a canonical continuum-time OS semigroup transported through the full sharp-local state.

Dependency summary.

Inputs. Theorem VI.1, the bounded-state restriction compatibility isolated in the Companion II detailed proof as Source Corollary 5.75, the finite-cap OS structure of Theorem V.10, and the inductive-union passage Corollary V.11

Imported standard background. the standard OS reconstruction theorem for reflection-positive states, applied only after the Lane A seam package has produced the full sharp-local positive-time state

Conclusion at this node. the bounded base state is identified inside the full sharp-local OS reconstruction, and its dyadic matrix elements extend to the canonical continuum-time semigroup for all $t \geq 0$; this theorem does not yet perform the density/graph-core or spectral-gap transfer

Used later in. Lemma VI.7 and Theorem VI.9

Remark VI.3 (QE3 transport readout rule). Once Theorem VI.2 is in hand, the QE3 route no longer changes generators or state spaces: the same OS Hamiltonian H appearing there is the operator whose spectrum is read in Theorem VI.9. The remaining passage to the full sharp-local Hilbert space uses only the density / finite-graph-norm package of Theorems VI.6 and VI.7. Thus no loss of spectral information can occur through a second reconstruction step or a hidden domain completion at the QE3 seam. After Theorem VI.2, the only imported black-box tool still used is the spectral theorem at the final semigroup-to-gap step.

Remark VI.4 (Dyadic-to-all-time route on the public spine). Theorem VI.1 is only the bounded dyadic OS-limit theorem on A_+ . The present theorem is the only public-spine step that upgrades those dyadic matrix elements to all $t \geq 0$, and it does so only after the capwise OS structure of Theorem V.10 and the inductive-union passage Corollary V.11 have produced the sharp-local OS semigroup. No direct lattice all-time theorem is assumed or needed.

Remark VI.5 (QE3 domain stability between semigroup and gap readout). The operator H appearing in Theorems VI.2 and VI.9 is fixed once the OS reconstruction of the full

sharp-local state has been formed. The spectral theorem in Theorem VI.9 is applied to this same self-adjoint operator on its native domain $\text{Dom}(H)$. The later density / graph-core package does not replace H , enlarge $\mathcal{H}$, or introduce an additional closed operator: it only approximates vectors already lying in the reconstructed Hilbert space, first in Hilbert norm and, on each fixed engineering cap, in the finitely many graph norms attached to that cap.

More explicitly, the QE3 passage has exactly the following order. First, Theorem VI.2 identifies one fixed reflection-positive sharp-local state and hence one fixed OS Hilbert space $\mathcal{H}$ together with one fixed self-adjoint Hamiltonian H . Second, Corollary VI.6 and Lemma VI.7 replace target vectors in $\Omega^\perp$ by approximants from the bounded positive-time / finite-cap core already sitting inside that same $\mathcal{H}$; these approximants are not new states and do not come from a second reconstruction. Third, the semigroup estimate is proved on that common core and then extended by Hilbert-space continuity, using $\|e^{-tH}\| \leq 1$, to the closure in the same $\mathcal{H}$. The graph-norm part of Lemma VI.7 controls only the legitimacy of applying the already constructed sharp-local operators on one fixed cap at a time; it is not a second closure procedure for H and it does not alter $\text{Spec}(H)$. Thus no new states, no enlarged Hamiltonian domain, and no spectrum-changing completion step enter between the continuum-time semigroup statement and the final gap readout.

D. Density and graph-core handoff

Corollary VI.6 (Density of the bounded base algebra). *In the setting of Theorem VI.2, let $(H, \Omega, \mathcal{H})$ be the OS reconstruction of the full sharp-local state. Then the image of A_+ is dense in $\mathcal{H}$, and*

$$\overline{\text{span}}\{[F - \omega_{\text{sharp}}(F)] : F \in A_+\} = \Omega^\perp.$$

In particular, the non-collapse argument may be carried out on bounded positive-time observables and extended by density to the full sharp-local Hilbert space.

Proof. This is exactly the bounded-base cyclicity export of Theorem V.14, applied to the full sharp-local state singled out by Theorem VI.2. $\square$

Lemma VI.7 (Graph-core approximation at the QE3 seam). *In the setting of Theorem VI.2, two approximation facts enter the final gap transfer.*

- (i) *By Corollary VI.6, the bounded positive-time algebra A_+ is dense in $\mathcal{H}$, and its centered classes are dense in $\Omega^\perp$.*
- (ii) *Whenever a finite family of renormalized sharp-local fields belongs to a fixed engineering cap $\Delta_{\max}$, the corresponding field-polynomial vectors are realized as limits of vectors from the common invariant core supplied by the finite-cap Lane A construction; the same approximants converge not only in the OS Hilbert norm but also in every finite graph norm relevant to that cap.*

For each $t \geq 0$ one has $\|e^{-tH}\| \leq 1$, so any semigroup estimate established first on the bounded positive-time core extends by Hilbert-norm continuity to the capwise sharp-local limits. Thus the QE3 passage from bounded observables to the full sharp-local Hilbert space uses only this explicit density/core package and no hidden domain argument.

Remark VI.8 (Visible finiteness of the graph-core package). Every sharp-local vector later used in the QE3 transport first lies in some fixed engineering cap, and only the finite list of graph norms attached to that cap enters the approximation argument. Lemma VI.7 therefore does not invoke a graph-norm completion uniform over the full inductive union; the full-union step enters only through density of A_+ and boundedness of the semigroup.

E. Continuum vacuum-gap transport

Theorem VI.9 (Continuum vacuum-gap transport). *Fix $u_{\text{ren}} \in (0, u_*]$ and let*

$$a_n := \ell_0 2^{-n}, \quad \beta_n := \beta(a_n; u_{\text{ren}})$$

be the tuned dyadic trajectory. Let ω_{sharp} be the full sharp-local state furnished by Corollary V.11, and let $(H, \Omega, \mathcal{H})$ be its OS reconstruction on the full sharp-local positive-time algebra. Assume there exists $m_ > 0$ such that*

$$\liminf_{n \rightarrow \infty} \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq m_* > 0.$$

Then the canonical continuum-time OS semigroup on the full sharp-local Hilbert space has spectral gap at least m_ ; equivalently,*

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty), \quad \ker(H) = \mathbb{C}\Omega.$$

The only topology used in the passage from bounded positive-time observables to sharp-local vectors is the Hilbert-norm / finite-graph-norm approximation package of Theorem VI.7. In particular, the constant m_ is read only from the tuned fixed-lattice lower bound and not from any capwise inverse-control or finite-cap graph-norm constant.*

Dependency summary.

Inputs. Proposition IV.1, Theorem IV.2, Theorem VI.1, Theorem VI.2, and Lemma VI.7

Imported standard background. the spectral theorem only, used through the local semigroup and Laplace-transform lemmas isolated in the Companion I detailed proof

Conclusion at this node. the tuned fixed-lattice lower bound transfers to the full sharp-local OS Hamiltonian on the vacuum sector; this is the gap theorem itself, not merely a clustering corollary or a later Minkowski reformulation

Used later in. Corollary VI.11, Theorem I.1, and the endpoint spectral corollaries in Section VII

Remark VI.10 (Decay-to-gap bridge at the QE3 readout). The last logical step is exactly the standard spectral-theorem bridge and nothing stronger. Once Theorems VI.2 and VI.7 place the bounded positive-time vectors and their sharp-local limits inside the same reconstructed Hilbert space $\mathcal{H}$, the QE3 chain yields the decay estimate in the exact operator form needed for the final spectral readout: for every $t \geq 0$,

$$\|e^{-tH}\|_{\Omega^\perp \rightarrow \Omega^\perp} \leq e^{-m_* t}.$$

Equivalently, for all $\xi, \eta \in \Omega^\perp$ one has

$$|\langle \xi, e^{-tH} \eta \rangle_{\mathcal{H}}| \leq e^{-m_* t} \|\xi\|_{\mathcal{H}} \|\eta\|_{\mathcal{H}}.$$

Indeed, the diagonal decay bound is first proved on the common bounded positive-time / finite-cap core, extends to all of $\Omega^\perp$ by the density package of Theorem VI.7 together with the contractivity $\|e^{-tH}\| \leq 1$, and then yields the displayed sesquilinear estimate by polarization. For a nonnegative self-adjoint operator, the spectral theorem identifies the operator norm on $\Omega^\perp$ with the supremum of $e^{-t\lambda}$ over $\lambda \in \text{Spec}(H \upharpoonright \Omega^\perp)$; hence the above bound is equivalent to

$$\text{Spec}(H \upharpoonright \Omega^\perp) \subset [m_*, \infty),$$

which is exactly the sharp-local vacuum-gap conclusion. Thus no additional Tauberian theorem, no second decay argument, and no further transport estimate intervene between QE3 decay and the final spectral-gap statement.

Corollary VI.11 (Continuum sharp-local vacuum gap). *Assume Proposition IV.1, and let ω_{sharp} be the full sharp-local state furnished by Corollary V.11 over the same tuned dyadic bounded base state. Let $(H, \Omega, \mathcal{H})$ be its OS reconstruction. Then*

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty), \quad \ker(H) = \mathbb{C}\Omega, \quad P_{(0, m_*)} = 0.$$

Equivalently,

$$\|e^{-tH}\psi\| \leq e^{-m_*t}\|\psi\| \quad (t \geq 0, \psi \perp \Omega).$$

This is the continuum sharp-local spectral consequence used later in Theorem I.1(D).

Proof. By Proposition IV.1, the tuned fixed-lattice lower bound satisfies

$$\liminf_{n \rightarrow \infty} \frac{\Delta_{\text{lat}}(\beta_n)}{a_n} \geq m_*.$$

Apply Theorem VI.9 with this value of m_* . The semigroup estimate is then the spectral-theorem reformulation of the same gap statement on $\Omega^\perp$. $\square$

The vacuum gap is now established on the continuum sharp-local vacuum sector. No later spectral-calculus wrapper is needed to prove it. Any downstream clustering or Hamiltonian reformulation may be cited as a consequence, but Theorems VI.9 and VI.11 are the point at which the proof of the gap itself is complete.

VII. ENDPOINT THEOREMS: RECONSTRUCTION, NON-TRIVIALITY, AND LOCAL-NET UNIVERSALITY

The preceding sections establish the sharp-local state and the continuum vacuum gap. This section collects the claim-facing consequences: the Euclidean OS structure on the full sharp-local algebra, the Wightman–Haag–Kastler reconstruction, field correspondence for the renormalized local gauge-invariant operators produced by Lane A, a non-triviality witness, and the faithful-Wilson universality theorem that identifies the exact local-net endpoint of the paper. The only black-box framework inputs used here are exactly those isolated in Section IB; no additional endpoint-side import enters beyond that list.

A. Reconstruction and non-triviality results

Theorem VII.1 (Euclidean OS structure on the full sharp-local algebra). *Let ω_{sharp} be the full sharp-local state of Corollary V.11. Then the Schwinger distributions generated by the*

renormalized local fields of Lane A satisfy the following properties on the full sharp-local algebra $\mathcal{A}_{\text{loc}}$:

- (i) *Euclidean invariance under translations and rotations.*
- (ii) *Bosonic symmetry under permutation of insertions.*
- (iii) *Osterwalder–Schrader reflection positivity on the full positive-time sharp-local algebra.*
- (iv) *Regularity and temperedness of the Schwinger distributions.*
- (v) *OS reconstruction on the full positive-time sharp-local algebra, yielding a Hilbert space*

$$(\mathcal{H}_{\text{loc}}, \Omega_{\text{loc}}, H_{\text{loc}}),$$

with $\Omega_{\text{loc}} = [1]$ cyclic for the full sharp-local positive-time algebra. Moreover the bounded positive-time base algebra A_+ embeds densely in $\mathcal{H}_{\text{loc}}$.

Theorem VII.2 (Wightman–Haag–Kastler reconstruction). *Let ω_{sharp} be as above. Then the Euclidean Schwinger functions of the full sharp-local theory admit a Minkowski reconstruction with the following properties:*

- (i) *there exists a Hilbert space $\mathcal{H}_{\text{loc}}$, a vacuum vector Ω_{loc} , and a strongly continuous unitary representation U of the Poincaré group whose joint spectrum lies in the closed forward light cone;*
- (ii) *for every renormalized local gauge-invariant field generated by Lane A there is a corresponding operator-valued tempered distribution on Minkowski space whose Wightman distributions are the analytic continuations of the Euclidean Schwinger distributions;*
- (iii) *these fields satisfy locality and generate the vacuum Haag–Kastler net of the local gauge-invariant sharp-local theory;*
- (iv) *the reconstruction is unique up to unitary equivalence once the Euclidean Schwinger functions are fixed.*

Corollary VII.3 (Field correspondence). *For every exact-dimension quotient label $[P]$ and every test function f , let*

$$O_{[P]}^{\text{ren}}(f; \mu)$$

denote the corresponding renormalized sharp-local field constructed by Lane A. Then in the Wightman reconstruction of Theorem VII.2 there exists a unique operator-valued tempered distribution $\Phi_{[P]}$ whose Euclidean Schwinger distributions are the analytic continuations of the mixed Schwinger functions of $O_{[P]}^{\text{ren}}(f; \mu)$. This correspondence is compatible with finite linear combinations, derivatives, quotient relations, finite engineering truncations, and the inductive-union passage to the full sharp-local algebra.

Consequently the reconstructed local field content is exactly the family of gauge-invariant local differential polynomials in the curvature and its covariant derivatives represented by the Lane A sharp-local algebra.

Proof. This is the field-correspondence output of Theorem VII.2 applied to the canonical exact-dimension quotient blocks and the inductive-union sharp-local algebra already constructed in Section V. $\square$

Corollary VII.4 (Minkowski Hamiltonian inherits the gap). *Let P^0 be the Minkowski Hamiltonian in the Wightman reconstruction of Theorem VII.2. Assume the continuum sharp-local vacuum-gap theorem Corollary VI.11. Then*

$$\text{Spec}(P^0) \subset \{0\} \cup [m_*, \infty).$$

In particular, the reconstructed local gauge-invariant sharp-local theory has positive energy, a unique vacuum, and a strictly positive Minkowski mass gap in its vacuum sector.

Proof. Under OS reconstruction, Euclidean time translations analytically continue to the Minkowski time-translation group. The Minkowski Hamiltonian therefore coincides with the OS generator on the reconstructed Hilbert space, so the spectral inclusion transfers directly from Corollary VI.11. $\square$

Theorem VII.5 (Non-triviality witness). *Assume the full sharp-local OS/Wightman construction of Theorem VII.2 together with the Route 1 vacuum-gap theorem Corollary VI.11. Let E^{ren} be the canonical centered CP-even dimension-4 scalar witness produced by the Lane A normalization scheme, and let $\varphi_{*,S}$ be its positive-time reference test function. Then:*

(i)

$$\|E^{\text{ren}}(\varphi_{*,S})\Omega_{\text{loc}}\|^2 = 1;$$

(ii) *the centered Euclidean and Minkowski two-point functions of E^{ren} are therefore nonzero;*

(iii) the spectral measure of the nonzero vector $E^{\text{ren}}(\varphi_{*,S})\Omega_{\text{loc}}$ under both H_{loc} and the Minkowski Hamiltonian P^0 is a nonzero finite positive measure supported in $[m_*, \infty)$.

In particular, the reconstructed local gauge-invariant sharp-local theory is non-trivial in the load-bearing sense required later in the paper: it contains a local gauge-invariant field that is not a multiple of the identity and has nonzero vacuum-created spectral weight above the gap.

Downstream Euclidean clustering. The dense-core Euclidean clustering statement in Theorem I.1(B) is a downstream corollary of the vacuum-gap theorem on the same sharp-local state: once Corollary VI.11 is known, the standard OS semigroup estimate on a dense positive-time probe core gives exponential decay at rate m_* . Its full proof remains outside the numbered body because the gap itself has already been established in Section VI.

B. Faithful-Wilson universality and exact local-net endpoint

Theorem VII.6 (Faithful-Wilson universality hypotheses). *Let G be compact connected simple and let ρ_1, ρ_2 be faithful finite-dimensional unitary Wilson representations. Form the finite continuum-normalized action family built from the two faithful Wilson plaquette actions and impose the same admissible gradient-flow renormalization prescription. Then the non-collapse, one-shot entrance, and ultraviolet-matching hypotheses required by the universality comparison theorems hold uniformly on this faithful-Wilson comparison family.*

Equivalently, the finite faithful-Wilson comparison family meets the Route 1 universality hypotheses simultaneously for both faithful regularizations, each at its own entrance block scale, with common family-wise constants obtained by taking maxima over the two faithful choices.

Theorem VII.7 (Qualitative faithful-Wilson universality of the continuum local theory). *Let G be compact connected simple and let ρ_1, ρ_2 be faithful finite-dimensional unitary Wilson representations. Construct the corresponding continuum theories using the same admissible gradient-flow renormalization scheme and the same coherent normalization conditions. Then there exists a unique $*$ -algebra isomorphism*

$$\Phi_{\rho_1 \rightarrow \rho_2} : \mathcal{A}_{\text{loc}}(G, \rho_1) \xrightarrow{\cong} \mathcal{A}_{\text{loc}}(G, \rho_2)$$

which, on every finite engineering cap, sends the Lane A renormalized field attached to an exact-dimension quotient label $[P]$ and test function f to the renormalized field with the

same label $[P]$ and the same test function f . The induced map intertwines all finite-family Schwinger functions, the regionwise local nets, the OS vacuum reconstructions, and the reconstructed Wightman fields.

In particular, within the admissible gradient-flow renormalization class and after coherent normalization, the continuum local gauge-invariant sharp-local Yang–Mills theory and its vacuum gap depend only on G and not on the faithful Wilson representation used to regularize the ultraviolet lattice theory.

Corollary VII.8 (Quantitative certificate remains ρ -indexed). *Fix a compact connected simple group G and choose a faithful finite-dimensional unitary Wilson regularization ρ . Then Theorem VII.7 identifies all faithful choices with the same continuum local theory $\mathcal{A}_{\text{loc}}(G)$, but the explicit Component 1 / Route 1 certificate retained in this manuscript is still recorded through the ρ -indexed constants*

$$\beta_*(G, \rho), \quad q^*(G, \rho), \quad \text{Cent}(G, \rho), \quad m_{\text{blk}}(G, \rho), \quad m_*(G, \rho; u_{\text{ren}}).$$

The present manuscript does not claim that these explicit quantitative certificates are uniform over all faithful ρ . Different faithful Wilson regularizations yield canonically isomorphic continuum local theories, but may certify different explicit lower bounds on their common vacuum gap.

Proof. The first statement is Theorem VII.7. The second follows because the explicit constants are still read from the chosen faithful ultraviolet regularization through the ultraviolet gate, one-shot entrance, and continuum transport theorems of Sections III, IV, and VI. No representation-uniform quantitative comparison theorem for those constants is proved in the present manuscript. $\square$

Theorem VII.9 (Exact local-net endpoint and exclusion of extended-support sector data). *Fix a compact connected simple gauge group G . Choose any faithful finite-dimensional unitary Wilson representation ρ_{fid} and let*

$$\mathcal{A}_{\text{loc}}(G, \rho_{\text{fid}}), \quad (\mathcal{H}_{\text{loc}}(G, \rho_{\text{fid}}), \Omega_{\text{loc}}(G, \rho_{\text{fid}}), U_{G, \rho_{\text{fid}}})$$

be the sharp-local Euclidean/Wightman theory produced from that faithful ultraviolet regularization. By Theorem VII.7, different faithful choices of ρ_{fid} yield canonically isomorphic theories, so we may write

$$\mathcal{A}_{\text{loc}}(G), \quad (\mathcal{H}_{\text{loc}}(G), \Omega_{\text{loc}}(G), U_G)$$

for that canonical faithful- ρ -independent local theory. Then:

- (i) by Corollary VII.3, the local Wightman fields in this theory are precisely the renormalized gauge-invariant local differential polynomials in the curvature and its covariant derivatives represented by Lane A ;
- (ii) if ρ is any finite-dimensional unitary Wilson representation of G and $G_{\text{eff}} := G/\ker(\rho)$, then the local observable net obtained from the ultraviolet pair (G, ρ) is canonically identified with $\mathcal{A}_{\text{loc}}(G_{\text{eff}})$; in particular, for faithful ρ this is just the canonical theory $\mathcal{A}_{\text{loc}}(G)$;
- (iii) line operators, surface defects, and other global-form-sensitive observables are not elements of the bounded-region local field algebra constructed here, because their localization regions are extended rather than bounded.

Consequently the theorem-level object constructed in this manuscript is the canonical local gauge-invariant sharp-local net $\mathcal{A}_{\text{loc}}(G)$ together with its OS/Wightman vacuum representation, and no extended-support nonlocal sector data enter that theorem-level existence-and-gap claim.

Corollary VII.10 (Group-only restatement of the endpoint theorem). *Theorem I.1 is a theorem about the local gauge-invariant Yang–Mills theory attached to each compact connected simple gauge group G . Any faithful finite-dimensional unitary Wilson representation may be used as an auxiliary ultraviolet regularization and, within the admissible gradient-flow renormalization class and after coherent normalization, yields the same continuum local theory $\mathcal{A}_{\text{loc}}(G)$ up to the canonical faithful-Wilson identification of Theorem VII.7. The explicit Route 1 certificate constants retained in Theorem I.1 remain indexed by the chosen faithful ultraviolet regularization, as recorded in Corollary VII.8. A nonfaithful ultraviolet choice yields instead the local theory of the effective quotient $G_{\text{eff}} := G/\ker(\rho)$. Differences between global forms survive only in nonlocal sector data not asserted by Theorem I.1; differences between faithful ultraviolet Wilson representations survive only through the present quantitative certificate, not through the continuum local theory itself.*

Proof. This is immediate from Theorems VII.7 and VII.9 together with Corollary VII.8. $\square$

Together these results fix the theorem-level object proved in this paper: a local gauge-invariant sharp-local Yang–Mills theory whose vacuum Hamiltonian has a spectral gap and whose continuum local net is independent of the faithful ultraviolet Wilson regularization.

What remains is only to assemble Theorem I.1 from the component decomposition already stated.

Endpoint claim hygiene. Within the admissible gradient-flow renormalization class and after coherent normalization, the endpoint results prove exactly one theorem-level object: the canonical bounded-region local gauge-invariant sharp-local net $\mathcal{A}_{\text{loc}}(G)$ together with its OS/Wightman vacuum representation and vacuum-sector mass gap. The faithful-Wilson universality theorem is qualitative only; Corollary VII.8 records separately that the explicit Route 1 quantitative certificate remains ρ -indexed. Extended-support line and surface sectors remain outside the theorem-level claim by Theorem VII.9.

VIII. ASSEMBLY OF THE MAIN THEOREM

We now assemble Theorem I.1 from the component decomposition stated in Sections III–VII. No new analytic input enters here. Each part of the theorem follows by concatenating the already named ultraviolet, Route 1, Lane A, QE3, and endpoint theorem packages in the order fixed by the public proof spine.

Proof of Theorem I.1. Part (A). The ultraviolet/public-scope gate of Section III furnishes the compact-simple A1 theorem Theorem III.1, the public group-scope export Corollary III.2, and the one-shot entrance theorem Proposition III.3. Those three results are exactly the theorem-level input required for the one-shot entrance statement in Theorem I.1(A). The faithful- ρ scope caveat and the effective-group boundary were fixed already in Section I and are preserved unchanged.

Part (B). The Euclidean clustering statement in Theorem I.1(B) is downstream of the vacuum-gap chain. Once the continuum sharp-local gap of Corollary VI.11 is known, the standard OS semigroup estimate on a dense positive-time probe core yields exponential decay at rate m_* . This is a consequence of the same sharp-local OS state used throughout the paper and does not add a second gap proof.

Part (C). Lane A closes the constructive route to the full sharp-local algebra through the flowed-state and bounded-base pair Theorems V.1 and V.2, the exact-dimension quotient and finite-truncation results Theorems V.3 to V.7, the mixed-channel finite-cap closure node Theorem V.9, the finite-cap sharp-local extension Theorem V.10, the inductive-union passage Corollary V.11, and bounded-base cyclicity Theorem V.14. The endpoint theorems

of Section VII then yield the Euclidean OS structure Theorem VII.1, the Wightman–Haag–Kastler reconstruction Theorem VII.2, the field correspondence Corollary VII.3, and the non-triviality witness Theorem VII.5. Together these give Theorem I.1(C).

Part (D). Route 1 produces the tuned fixed-lattice OS gap Proposition IV.1, returns it to the physical Wilson theory through Theorem IV.2, and seals the weak-window intake with the sole live certificate Theorem IV.4. The final transport chain is then exactly the bounded Wilson limit Theorem VI.1, the continuum-time OS upgrade Theorem VI.2, the density and graph-core handoff Corollary VI.6 and Lemma VI.7, the unique transport node Theorem VI.9, and the continuum sharp-local vacuum-gap corollary Corollary VI.11. This proves Theorem I.1(D).

Finally, the theorem-level endpoint and the faithful-Wilson caveat are fixed by Theorems VII.6 to VII.10. Thus the theorem proved in Theorem I.1 is exactly a theorem about the bounded-region local gauge-invariant sharp-local net attached to the compact connected simple group G , with faithful-Wilson universality at the level of the continuum local theory and a still- ρ -indexed explicit Route 1 quantitative statement. $\square$

This completes the proof of Theorem I.1. The core paper has now done its intended job: it has exposed the claim, the exact theorem-level endpoint, the finite public component decomposition, and the principal interface statements, while the detailed analytic and constructive proofs live in stable companion papers.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

Appendix A: Theorem map and dependency DAG

This appendix is purely navigational. It records the theorem-level dependency map used in the core paper and adds no analytic input. Only essential edges are displayed; auxiliary reformulations and illustrative side branches are omitted.

Theorem components in dependency order

Component 1.:

Theorem N.20; Corollary N.21. Compact-simple A1 ultraviolet gate and public group-scope export. Primary detailed proof: Core Section 3 / Companion I.

Component 2.:

Proposition F.3, with supporting results Theorem F.317 and Corollary F.318. One-shot entrance into the KP/Dobrushin basin at bounded physical scale. Primary detailed proof: Core Section 3 / Companion I.

Component 3.:

Proposition F.4, with proof-local culmination at Theorem F.308. Tuned full fixed-lattice OS gap. Primary detailed proof: Core Section 4 / Companion I.

Component 4.:

Theorem 4.92; Corollaries 4.93–4.95. Unique flowed continuum state, Wilson/RG identification on $\mathcal{A}_{\text{flow}}$, tuned bounded positive-time base state, and dyadic reflected OS limits. Primary detailed proof: Core Sections 5–6 / Companion II, Appendix A, with the flowed de-patching seam in Companion I, Section III.

Component 5.:

Theorem L.8; Theorem L.11; Theorem 5.39; Theorem 5.69; Corollary F.245. Exact-dimension breadth plus finite-truncation inverse/remainder control. Primary detailed proof: Core Section 5 / Companion II.

Component 6.:

Corollary F.330A; Theorem F.331; promoted bridge 5.74A; Theorem 5.74; Corollary 5.75; Corollary 5.76. First mixed-correlator bridge at finite cap, theorem-level positive/unital bridge into 5.74, finite-cap sharp-local extension, bounded-state compatibility, then extension to the full sharp-local inductive union. Primary detailed proof: Core Sections 5–6 / Companion II.

Component 7.:

Theorem 5.77. Bounded-base cyclicity. Primary detailed proof: Core Section 5 / Companion II.

Component 8.:

Core Theorem IV.2 (detailed proof: Companion I, Theorem III.18); Theorem F.211; Theorem F.212; Corollary F.213; Lemma F.214; Theorem F.216; Corollary F.5. Wilson-to-continuum transport and continuum vacuum gap. Component 8 explicitly consumes Component 4 through the tuned bounded base state, then the Lane A seam nodes Theorem 5.74(b), Corollary 5.75, Corollary 5.76, Theorem 5.77, Corollary F.213, and Lemma F.214. Primary detailed proof: Core Sections 4 and 6 / Companion I.

Component 9.:

Theorem M.1; Corollary M.2; Theorem M.3; Corollaries M.4–M.5; Theorem M.6. Downstream Euclidean clustering reformulation, OS/Wightman existence, field correspondence, Minkowski-gap transfer, and non-triviality. Primary detailed proof: Core Section 7 / Companion III.

Component 10.:

Theorem O.2; Theorem O.3; Corollary O.4; Theorem O.5; Corollary O.7. Qualitative faithful-Wilson universality, the ρ -indexed quantitative-status note, and the exact theorem-level local-net endpoint. Primary detailed proof: Core Section 7 / Companion III.

Auxiliary node.:

Theorem F.298. Sole live weak-window sufficiency certificate on the public Route 1 spine.

Theorem-level dependency DAG**Dependency summary****Route 1 chain.:**

Component 1 $\rightarrow$ Component 2 $\rightarrow$ Component 3 $\rightarrow$ Component 8 $\rightarrow$ Component 9 $\rightarrow$ Component 10.

Lane A chain.:

Component 4 $\rightarrow$ Component 5 $\rightarrow$ Component 6 $\rightarrow$ Component 7, with Component 4 feeding Component 8 through the bounded base state, Component 6 and Component 7

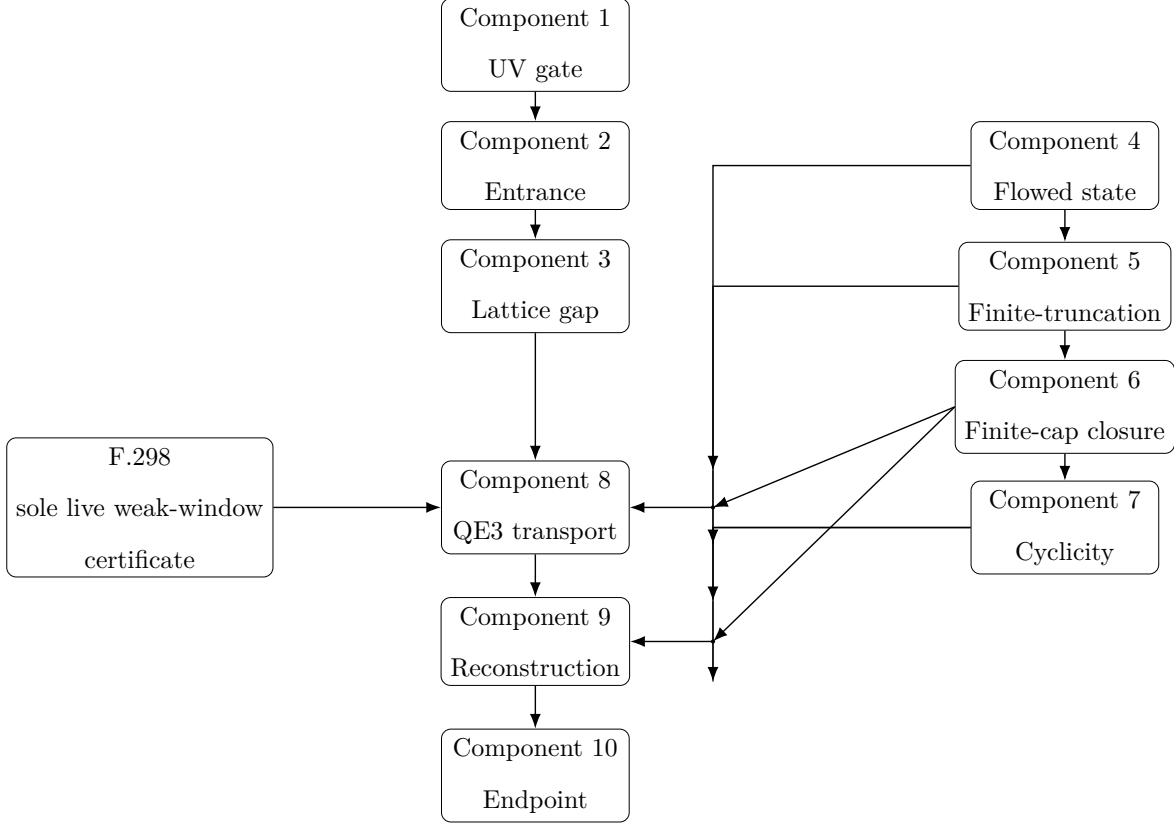


FIG. 2. Theorem-level dependency DAG. Component 8 reuses Component 4 through the bounded positive-time base state and receives the Lane A interface outputs. Theorem F.298 is the sole weak-window certificate on the public spine; Lane B is omitted.

feeding Component 8 through the repaired QE3 seam, and Components 5, 6, 7, and 8 feeding Component 9 through field correspondence, inductive-union / cyclicity, and the downstream clustering-gap reformulations.

Sole auxiliary certificate.:

Theorem F.298 feeds Component 8 only.

Explicit exclusions.:

Lane B, archival Route 1 branches, illustrative classical-family appendices, and long constant tables carry no theorem-level edge in the core paper.

Together with the master index and numbering crosswalk in Appendix C, this appendix records the dependency surface used in the submission.

Appendix B: Imported background and companion concordance

This appendix records the small amount of standard background used as black-box input in the core paper and states where the detailed proofs of the companion results are located.

Imported standard background

KP/Dobrushin entrance-to-clustering package.:

Used in Sections III and IV, and in the downstream clustering clause of Section VIII. It supplies the standard clustering mechanism once one-shot entrance has been established; it is not a substitute for the Route 1 transport chain.

Standard OS reconstruction / analytic continuation / uniqueness theorem family.:

Used in Section VII, especially Theorems VII.1, VII.2 and VII.7. It upgrades the constructive Euclidean OS structure to the Wightman/Haag–Kastler reconstruction and provides the uniqueness clause used in faithful-Wilson universality.

Spectral theorem and semigroup calculus.:

Used in Section VI and in the endpoint discussion only after the sharp-local semigroup and Hamiltonian have already been constructed theorem-locally.

Standard Wilson-lattice reflection positivity.:

Used on the lattice side as the usual positive-time positivity input in the fixed-lattice and continuum-transport arguments.

Detailed-proof concordance

Core-only results.:

The framing results in Sections I–II and the final assembly in Section VIII are stated and proved in the core paper itself.

Companion I.:

Core results 3.1, 3.2, 3.3, 4.1, 4.2, 4.4, 6.1, 6.2, 6.4, 6.5, 6.7, and 6.8 route to Companion I, Sections 1–6 and Appendix F.

Companion II.:

Core results 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8, 5.9, 5.10, 5.11, and 5.13 route to Companion II, Sections 1–7 and Appendices A–E.

Companion III.:

Core results 7.1, 7.2, 7.3, 7.4, 7.5, 7.6, 7.7, 7.8, 7.9, and 7.10 route to Companion III, Sections 1–4 and Appendices M and O.

Material deferred from the core presentation

- Lane B and the spectral-calculus wrapper remain downstream reformulations and do not occupy numbered theorem space in the core paper.
- Kernel-parametrix detail, insertion-RG proofs, and long coefficient-matrix arguments remain in the companion detailed proofs.
- AQFT commentary on sectors and field-algebra extensions remains outside the theorem-level endpoint proved in the core paper.

This appendix is included only to keep the proof route transparent: every imported theorem family is named explicitly, and every detailed proof cited from the core paper has a unique companion reference.

Appendix C: Master theorem index and canonical numbering crosswalk

This appendix is purely navigational. It records one canonical label, one source-manuscript cross-reference, one component assignment, one detailed-proof reference, one theorem role, and one immediate-consumer list for every essential or endpoint-facing node on the public spine.

Canonical numbering rule. The core-paper labels are the public canonical labels. Companion/source labels are retained in the detailed-proof references and are frozen here only as cross-reference labels. The “Immediate consumers” column records theorem nodes and endpoint clauses on the live spine, not every narrative mention.

Component-level canonical numbering cross-reference

No.	Core	Source	Detailed proof	Role
1	III.1–III.2	N.20, N.21	Comp. I, §1	compact-simple ultraviolet gate and public group-scope export
2	III.3	F.3, with supporting results by F.317–F.318	Comp. I, Appendix F / §2	one-shot entrance and bounded physical entrance scale
3	IV.1	F.4 (QE2 _{lat} theorem), with supporting result at F.308	Comp. I, Appendix F / §3	tuned full fixed-lattice vacuum-gap lower bound
4	V.1–V.2	4.92–4.95	Comp. II, Appendix A (+ Comp. I, §3 for de-patching)	flowed continuum state, bounded base state, dyadic OS limits
5	V.3–V.7	L.8, L.11, 5.39, 5.69, F.245	Comp. II, §§2–3 / Appendices E, B, D	exact-dimension quotient control and finite-truncation inverse package

No.	Core	Source	Detailed proof	Role
6	V.8–V.11	F.330A, F.331, 5.74A, 5.74, 5.75, 5.76	Comp. II, §§4–6 / Appendices D, B	finite-cap mixed closure, theorem-level state bridge, capwise extension, bounded-state compatibility, inductive-union passage
7	V.13	5.77	Comp. II, §6 / Appendix B	bounded-base cyclicity
8	IV.2, VI.1, VI.2, VI.4, VI.5, VI.7, VI.8	III.18, F.211, F.212, F.213, F.214, F.216, F.5, with Lane A seam inputs 5.74(b), 5.75, 5.76, 5.77	Comp. I, §4 and Appendix F	same-scale Wilson return, QE3 transport, continuum vacuum gap
9	VII.1–VII.5	M.1, M.2, M.3, M.4, M.5, M.6	Comp. III, §§1–3	downstream Euclidean clustering reformulation, OS package, reconstruction, field correspondence, Minkowski-gap transfer, non-triviality
10	VII.6–VII.10	O.2, O.3, O.4, O.5, O.7	Comp. III, §4	qualitative faithful-Wilson universality, ρ -indexed quantitative status, and exact local-net endpoint
Cert.	IV.4	F.298	Comp. I, §5	sole live weak-window sufficiency certificate on the public spine

Crosswalk note. Component 8 is not Route 1-internal: it reuses Component 4 through the bounded base state 4.94 and explicitly consumes the Lane A seam nodes 5.74(b), 5.75, 5.76,

5.77, F.213, and F.214.

Master theorem index: front-end and Route 1 nodes

Canon.	Source	Comp.	Detailed proof	Role	Next use
I.1	—	—	Core §8	assembled theorem object and exact endpoint claim	—
II.1	—	—	Core §2	records the ten-part reduction and public theorem map	Appendix A; navigational only
III.1	N.20	1	Comp. I, §1 / Core §3	compact-simple A1 ultraviolet gate	III.2, III.3
III.2	N.21	1	Comp. I, §1 / Core §3	public group-scope export on the live spine	III.3, II.1
III.3	F.3; supporting results F.317–F.318	2	Comp. I, Appendix F / Core §3	one-shot entrance at bounded physical scale along the tuned dyadic path	IV.1, I.1(A)
IV.1	F.4; supporting result F.308	3	Comp. I, Appendix F / Core §4	tuned full fixed-lattice gap normalized at the physical scale later used in QE3	VI.7, VI.8, I.1(D)
IV.2	III.18	8	Comp. I, §4 / Core §4	same-scale Wilson-to-patched return on the logarithmic dyadic depths used downstream	VI.1, VI.7
IV.4	F.298	cert.	Comp. I, §5 / Core §4	sole live weak-window sufficiency certificate	Route 1 summary of VI.7; no later theorem strengthens it

Master theorem index: Lane A construction nodes

Canon.	Source	Comp.	Detailed proof	Role	Next use
V.1	4.92 + 4.93	4	Comp. II, §1 / Appendix A (Comp. I, §3)	unique flowed continuum (+ state and executive Wil-son/RG identification on $\mathcal{A}_{\text{flow}}$)	V.8, V.10
V.2	4.94	4	Comp. II, §1 / Appendix A	tuned bounded positive-time base state	V.8, V.10, VI.1, V.13
V.3	L.8	5	Comp. II, §2 / Appendix E	exact-dimension quotient identity on symmetry blocks	V.4, V.5
V.4	L.11	5	Comp. II, §2 / Appendix E	coefficient extraction on arbitrary quotient blocks	V.5, V.6
V.5	5.39	5	Comp. II, §3 / Appendix B	one-shell transport on finite truncations	V.6
V.6	5.69	5	Comp. II, §3 / Appendix B	finite-truncation inverse control	V.7, V.9, V.10
V.7	F.245	5	Comp. II, §3 / Appendix D	capwise SFTE/remainder compatibility package	V.9, V.10
V.8	F.330A	6	Comp. II, §4 / Appendix D	finite mixed bounded-family packaging at fixed cap	V.9
V.9	F.331	6	Comp. II, §4 / Appendix D	first mixed-correlator closure node at finite cap	V.10
V.10	5.74	6	Comp. II, §5 / Appendix B	finite-cap sharp-local extension with capwise OS structure on the capwise algebra; theorem-level state input supplied by 5.74A	V.11, VI.2
V.11	5.76	6	Comp. II, §6 / Appendix B	passage from finite caps to the full sharp-local inductive union	VI.2, VII.1
V.13	5.77	7	Comp. II, §6 / Appendix B	bounded-base cyclicity in the reconstructed sharp-local Hilbert space	VI.4, VII.1

Master theorem index: QE3 seam and continuum-gap transport nodes

Canon.	Source	Comp.	Detailed proof	Role	Next use
VI.1	F.211	8	Comp. I, Ap- pendix F / §6	bounded Wilson/dyadic OS limit theorem along the tuned trajectory	VI.2, VI.7
VI.2	F.212	8	Comp. I, Ap- pendix F / §6	continuum-time OS upgrade of the bounded base state via Lane A, with the local role split 5.75 / 5.74(b) / 5.76	VI.5, VI.7
VI.4	F.213	8	Comp. I, Ap- pendix F / §6	density of the bounded base algebra in the full sharp-local Hilbert space	VI.5, VI.7
VI.5	F.214	8	Comp. I, Ap- pendix F / §6	explicit Hilbert/graph-core approximation package at the QE3 seam	VI.7
VI.7	F.216	8	Comp. I, Ap- pendix F / §6	unique continuum vacuum- gap transport node	VI.8, VII.4, VII.5, I.1(D)
VI.8	F.5	8	Comp. I, Ap- pendix F / §6	continuum sharp-local vacuum-gap corollary	VII.4, VII.5, I.1(B,D)

Master theorem index: endpoint and universality nodes

Canon.	Source	Comp.	Detailed proof	Role	Next use
VII.1	M.1	9	Comp. III, §1 / Core §7	Euclidean OS package on the full sharp-local algebra	VII.2, I.1(C)
VII.2	M.3	9	Comp. III, §2 / Core §7	Wightman–Haag–Kastler re-construction	VII.3, VII.4, VII.5, VII.9
VII.3	M.4	9	Comp. III, §2 / Core §7	field correspondence for the Lane A local fields	VII.9
VII.4	M.5	9	Comp. III, §2 / Core §7	Minkowski Hamiltonian inherits the Route 1 gap	endpoint spectral consequence only
VII.5	M.6	9	Comp. III, §3 / Core §7	non-triviality witness and nonzero spectral weight above the gap	I.1(C)
VII.6	O.2	10	Comp. III, §4 / Core §7	faithful-Wilson comparison family satisfies universality hypotheses uniformly	VII.7
VII.7	O.3	10	Comp. III, §4 / Core §7	qualitative faithful-Wilson universality of the continuum local theory	VII.8, VII.9, VII.10
VII.8	O.4	10	Comp. III, §4 / Core §7	explicit quantitative state-ment remains ρ -indexed	VII.10, I.1 endpoint caveat
VII.9	O.5	10	Comp. III, §4 / Core §7	exact local-net endpoint and exclusion of extended-support sectors	VII.10, I.1 theorem object
VII.10	O.7	10	Comp. III, §4 / Core §7	group-only restatement of the exact theorem endpoint	I.1 final framing

Companion-only live seam nodes without separate core numbers

The source labels III.18, F.211, F.212, F.213, F.214, F.216, F.298, F.331, 5.74, 5.76, 5.77, M.1, M.2, M.3, M.6, O.2, O.3, O.5, and O.7 already appear in the source-label column of the master-index tables above. The additional companion-only seam nodes below remain canonical inside their detailed proofs and are listed separately because they have no separate core-paper numbers.

Companion label	Comp.	Detailed proof	Role	Next use
5.74A	6	Comp. II, §5 / Appendix B	theorem-level positive/unital finite-cap bridge into Theorem 5.74; isolates exactly the capwise state/positivity input and nothing beyond it	5.74 / V.10; Component 6 seam summary
5.75	6	Comp. II, §5 / Appendix B	bounded-state restriction compatibility at the finite-cap seam; this is <i>not</i> the inductive-union theorem	F.212 / VI.2; Component 8 seam summary
F.317	2	Comp. I, Appendix F	theorem-level one-shot entrance theorem inside the QE1 detailed proof	F.318, F.3 / III.3
F.318	2	Comp. I, Appendix F	bounded physical one-shot entrance scale along the tuned dyadic trajectory	F.3 / III.3, F.308 / IV.1
F.308	3	Comp. I, Appendix F	full fixed-lattice OS gap from one-shot entrance, used to prove the QE2 _{lat} theorem	F.4 / IV.1
M.2	9	Comp. III, Appendix M	downstream positive-time Euclidean clustering and vacuum-uniqueness reformulation of the already proved gap	I.1(B); Component 9 summary

Taken together, the crosswalk and index tables in this appendix answer three navigational questions for every theorem number on the public spine: where is it proved, what does it do, and what consumes it next.

Appendix D: Lean formalization boundary

This appendix records the scope of the accompanying Lean 4 formal companion. It adds no mathematics and does not enlarge the theorem claim. Its role is only to identify what is formalized natively, what remains outside that native scope, and which standard framework inputs are still used as black-box imports.

Current native scope

At the current stage, the Lean companion formalizes the manuscript-facing theorem surface rather than the full analytic Yang–Mills proof corpus. Concretely, the formal layer encodes:

- (i) the theorem surface and companion ownership map;
- (ii) the ten-part decomposition, dependency spine, and decisive seam nodes used on the public proof spine;
- (iii) the theorem/source crosswalk and named witness exports for the constructive, vacuum-gap, and endpoint chains; and
- (iv) one full manuscript-facing native development that assembles those theorem-level objects at a single axiom-free manuscript boundary.

The formal companion should therefore be read as an auxiliary verification layer for theorem-level organization, dependency locality, seam visibility, witness ownership, and claim boundaries.

What the formal companion does not claim

The present Lean scope is *not* a claim that every analytic proof in Companions I–III has already been re-proved natively in Lean. In particular, it does not claim:

- (a) a full end-to-end Lean formalization of the underlying Yang–Mills QFT / PDE mathematics;
- (b) native discharge of every analytic argument cited by the manuscript detailed proofs; or

(c) replacement of the companion proofs by a fully internal Lean derivation.

Thus the strongest accurate reading is: the manuscript-facing theorem-object layer is formalized axiom-free, while the deeper analytic arguments remain carried by the synchronized companion corpus.

Imported standard black-box inputs

The only standard framework inputs intentionally left outside the present Lean scope are the following.

(A) Osterwalder–Schrader reconstruction framework.:

This is used only after the Euclidean package has been established. It supplies the standard reconstruction passage from Euclidean Schwinger functions to the Hilbert-space/Hamiltonian setting together with the usual uniqueness clause.

(B) Spectral theorem.:

This is used at the operator-theoretic transport step: it supplies the semigroup calculus and the passage from the stated operator bounds to the spectral-gap conclusion.

(C) Standard Wilson-lattice reflection positivity.:

This is used as the standard lattice input furnishing positivity on the Wilson side before the OS reconstruction machinery is invoked.

No broader imported analytic package is hidden here: these are the only standard black-box inputs intentionally retained outside the current Lean scope.

Use of the formal companion

The Lean companion is supplied as an auxiliary verification aid, not as a replacement for the detailed proofs. It may be used to check the public decomposition, theorem-level organization, seam locality, and endpoint scope without first traversing the full companion corpus. The mathematical content remains the content of the core paper and Companions I–III; the Lean layer records one formal organization of that content at the theorem-level level.

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